Drinfeld's Moduli Theorem

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Abstract

In this report, we first recall the theory of formal groups and the Cartier–Dieudonné theory, which establishes an equivalence between the category of formal groups and a category of modules equipped with additional structures. This equivalence shows that a formal group can be described in terms of linear algebraic data. The main objects of interest are special formal $\mathcal{O}_D$-modules. Drinfeld's theorem shows that the p-adic upper half plane can be realized as a moduli space parametrizing these objects together with suitable level structure. The main reference for this report is [1]. For the details on formal groups and Cartier-Dieudonné theory, see [2]. For the details on étale morphisms and constructible sheaves, see [3].

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Notations

Let K be a nonarchimedean local field[1] with ring of integers $\mathcal{O}$, uniformizer π, and residue field $k = \mathcal{O}/\pi\mathcal{O}$ of cardinality q. Let K'/K be the unramified quadratic extension, with ring of integers $\mathcal{O}'$. Write σ for the nontrivial element of $\mathrm{Gal}(K'/K)$.

Let D be the quaternion division algebra over K with invariant $1/2$. Its maximal order has the presentation (which is all we need)

$$\mathcal{O}_D = \mathcal{O}'[\Pi], \qquad \Pi^2 = \pi, \qquad \Pi a = \sigma(a)\Pi \quad (a \in \mathcal{O}').$$

Unless otherwise stated, B denotes an $\mathcal{O}$-algebra.

1 Formal groups and formal modules

In the first three subsections, we could regard B simply as a ring.

1.1 Formal groups (as functors)

Let Nilp_B be the category of nilpotent, possibly nonunital, B-algebras. Equivalently, an object may be viewed as an augmented B-algebra $R \twoheadrightarrow B$ whose augmentation ideal is nilpotent. And the euivalence is given by:

$$\begin{aligned} \mathrm{Nilp}_B \cong \mathrm{AugAlg}_B &:= \{R \in \mathrm{Alg}_B \mid \exists \epsilon : R \to B \text{ s.t. } \ker \epsilon \text{ is nilpotent}\} \\ N &\mapsto B \oplus N \text{ where } \epsilon \text{ is the natural projection,} \\ \ker \epsilon &\hookleftarrow (R, \epsilon). \end{aligned}$$

Definition 1.1. A (commutative) formal group over B is an additive, left exact functor [2]

$$G : \mathrm{Nilp}_B \longrightarrow \mathbf{Ab}$$

which satisfies the usual formal smoothness condition, i.e. if $N_1 \twoheadrightarrow N_2$, then $G(N_1) \to G(N_2)$ is surjective.

Example 1.2. If

$$\mathbf{G} = (G_1, \ldots, G_n) \in B[\![X_1, \ldots, X_n, Y_1, \ldots, Y_n]\!]^n$$

is a commutative formal group law, then

$$G(N) = N^n, \qquad x +_G y = (G_1(x, y), \ldots, G_n(x, y))$$

defines a formal group. We denote it by $\mathrm{Spf}\, B[\![X_1, \ldots, X_n]\!]$.

Definition 1.3. If G is a formal group over B and B' is a B-algebra, then we define the base change of G by compositing the "forgettable functor" $\mathrm{Nilp}_{B'} \to \mathrm{Nilp}_B$.

Definition 1.4. The morphisms between formal groups are just natrual transformations.

Definition 1.5. For any morphsim $\varphi : G_1 \to G_2$ between two formal groups G_1, G_2 over B, the kernel of φ is $\ker \varphi : \mathrm{Nilp}_B \to \mathbf{Ab}$ given by sending N to $\ker(\varphi(N))$. It is also a formal group.

One easy observation is that kernels commute with base change:

$$(\ker \varphi)_{B'} \cong \ker(\varphi_{B'}).$$

[1] i.e. a finite extensiopn of $\mathbb{Q}_p$.

[2] We call a functor additive (resp. left exact) if it commutes with arbitrary direct sum (resp. finite limits).

1.2 The tangent space and the Lie algebra

Definition 1.6. Given a formal group G, we define its tangent space t_G as the functor $G \circ \iota$ where $\iota : \mathrm{Mod}_B \to \mathrm{Nilp}_B$ is the natural functor, since for any module M, we regard it as a square-zero B-algebra i.e. $M \cdot M = 0$.

Definition 1.7. Given a formal group G, we define its Lie algebra as

$$\mathrm{Lie}(G) = t_G(B) = G(B\varepsilon), \quad \varepsilon^2 = 0.$$

If $\mathrm{Lie}(G)$ is finite locally free, define

$$\dim G = \mathrm{rk}_B \mathrm{Lie}(G).$$

Addition and scalar multiplication on $B\varepsilon$ make $\mathrm{Lie}(G)$ a B-module.

Proposition 1.8. *There is a natural isomorphism*

$$t_G(M) \cong M \otimes_B \mathrm{Lie}(G).$$

Example 1.9. For $G = \mathrm{Spf}\, B[\![X_1, \ldots, X_n]\!]$, one has $\mathrm{Lie}(G) \cong B^{\oplus n}$ and then $\dim G = n$.

1.3 Representability and Prorepresentability

Definition 1.10. If R is a complete augmented B-algebra, we use $\mathrm{Spf}\, R$ to denote the functor sending N to $\mathrm{Hom}_{\mathrm{AugAlg_B}}(R, B \oplus N)$. We call a formal group G representable by R if G coincides with $\mathrm{Spf}\, R$ as set-valued functors.

Definition 1.11. Let R be a B-algebra. We call it a complete augmented B-algebra R if it admits a decreasing sequence of ideals

$$R \supset I_1 \supset I_2 \supset \cdots$$

such that $R \cong \varprojlim_\nu R/I_\nu$ and R/I_ν is an augmented B-algebra.[3] We also endow R a topology such that $\{I_\nu\}_\nu$ forms a neiborhood base of $0 \in R$. We use $\mathrm{Spf}\, R$ to denote the functor sending N to

$$\varinjlim_i \mathrm{Hom}_{\mathrm{AugAlg}_B}(R/I_i, B \oplus N).$$

We call a formal group G prorepresentable by R if G coincides with $\mathrm{Spf}\, R$ as set-valued functors.

Definition 1.12. A formal group G is called prorepresentable if there is a complete augmented B-algebra R such that $G \cong \mathrm{Spf}\, R$. Concretely, choose ideals

$$R \supset I_1 \supset I_2 \supset \cdots, \qquad R \cong \varprojlim_\nu R/I_\nu,$$

with every augmentation ideal of R/I_ν nilpotent. Then

$$G(N) = \varinjlim_\nu \mathrm{Hom}_{B,\mathrm{aug}}(R/I_\nu, B \oplus N).$$

Proposition 1.13. *If B' is a B-algebra and $G = \mathrm{Spf}\, R$ is a prorepresentable formal group over B, then the base change of G to B' is prorepresentable by the completed tensor product $R \widehat{\otimes}_B B'$.*

[3] i.e. R is a object in the pro category of $\mathrm{AugAlg_B}$.

Actually, most formal groups we are concerned are prorepresentable.

Proposition 1.14. *Suppose that G is a formal group. If $\mathrm{Lie}(G)$ is finite locally free over B, then G is prorepresentable. If $\mathrm{Lie}(G)$ is free of rank n, one may choose coordinates after which the representing algebra has the form $B[\![X_1, \ldots, X_n]\!]$.*

Remark 1.15. The finiteness of $\mathrm{Lie}(G)$ alone is not sufficient for an arbitrary functor. The exactness, additivity and formal smoothness hypotheses in the definition of a formal group are being used here.

1.4 Formal $\mathcal{O}$-modules and formal $\mathcal{O}_D$-modules

Definition 1.16. A formal $\mathcal{O}$-module over B is a formal group X with an $\mathcal{O}$-action, i.e. a homomorphism

$$i : \mathcal{O} \longrightarrow \mathrm{End}(X)$$

whose induced action on $\mathrm{Lie}(X)$ agrees with the structural action through $\mathcal{O} \to B$.

Now we require B to be an $\mathcal{O}'$-algebra.

Definition 1.17. A formal $\mathcal{O}_D$-module is a formal $\mathcal{O}$-module whose action extends to $i : \mathcal{O}_D \to \mathrm{End}(X)$. It is *special* if

$$\mathrm{Lie}(X) = \mathrm{Lie}(X)_0 \oplus \mathrm{Lie}(X)_1$$

under the two embeddings of $\mathcal{O}'$, and both summands are free B-modules of rank 1.

Explicitly,

$$\mathrm{Lie}(X)_j = \{m \in \mathrm{Lie}(X) \mid i(a)m = \sigma^j(a)m \ \forall a \in \mathcal{O}'\}, \quad j = 0, 1,$$

where a and $\sigma(a)$ acts on m through the B-module structure of $\mathrm{Lie}\, X$. The operator Π interchanges the two pieces. Since the $\mathcal{O}_D$-action is determined by $\mathcal{O}'$ and Π, we would see that it is somehow equivalent to giving an grading structure together with a degree-1 operator satisfying $\Pi^2 = \pi$ and $\Pi a = \sigma(a)\Pi$.

One may see the decomposition clearly by the following proposition.

Proposition 1.18. *Let M be a $\mathcal{O}'$-module with another $\mathcal{O}'$-action $i : \mathcal{O}' \to \mathrm{End}_{\mathcal{O}'}(M)$ (which is a homomorphism of $\mathcal{O}$-algebras). Then we have $M = M_0 \oplus M_1$ where*

$$M_j = \{m \in M \mid i(a)m = \sigma^j(a)m \ \forall a \in \mathcal{O}'\}, \quad j = 0, 1.$$

Proof. Choose $u \in \mathcal{O}^\times$ which is not a square and write $\mathcal{O}' = \mathcal{O}[\sqrt{u}]$ (or $\mathcal{O}[\frac{1+\sqrt{u}}{2}]$ depending on p odd or not). Then we have two projectors

$$P_0 = \frac{i(\sqrt{u}) + \sqrt{u}}{2\sqrt{u}}, \qquad P_1 = \frac{-i(\sqrt{u}) + \sqrt{u}}{2\sqrt{u}}.$$

They satisfy

$$P_0 P_1 = 0, \qquad P_0 + P_1 = 1, \qquad M_0 = \mathrm{im}(P_0), \quad M_1 = \mathrm{im}(P_1).$$

□

2 A brief introduction to Cartier–Dieudonné theory

2.1 Ramified Witt vectors

In this subsection, we will define the ramified Witt-vector ring. Let B be an $\mathcal{O}$-algebra as usual.

Definition 2.1. The underlying set of the ramified Witt-vector ring is $W_{\mathcal{O}}(B) = B^{\mathbb{N}}$. Its ring structure is characterized by the ghost map

$$w = (w_0, w_1, \cdots) : W_{\mathcal{O}}(B) \to B^{\mathbb{N}}$$

where

$$w_n(a_0, a_1, \ldots) = a_0^{q^n} + \pi a_1^{q^{n-1}} + \cdots + \pi^n a_n.$$

and it is an $\mathcal{O}$-algebra homomorphism. When B is π-torsion-free, the ghost map embeds $W_{\mathcal{O}}(B)$ into $B^{\mathbb{N}}$, so we can regard $W_{\mathcal{O}}(B)$ as a subalgebra of $B^{\mathbb{N}}$ which gives a well-defined algebra structure of $W_{\mathcal{O}}(B)$. In general, we choose a surjection $\tilde{B} := \mathcal{O}[X_b \mid b \in B] \twoheadrightarrow B$ given by sending X_b to b. Then $W_{\mathcal{O}}(\tilde{B}) \twoheadrightarrow W(B)$ gives the $\mathcal{O}$-algebra structure of $W_{\mathcal{O}}(B)$.

Remark 2.2. To verify that the above construction is well-defined, it remains to show that, in the π-torsion-free case, the image of the ghost map is an $\mathcal{O}$-subalgebra of $B^{\mathbb{N}}$, and that, in the general case, the induced operations are independent of the choices of lifts. We omit these verifications and point out that there is a more canonical construction of $W_{\mathcal{O}}$ using the Witt polynomials, which makes $W_{\mathcal{O}}$ naturally a functor.

We will then define two important operators on $W_{\mathcal{O}}(B)$, called the Verschiebung and Frobenius operators. They are often denoted by V and F in the literature. However, in Cartier–Dieudonné theory, these symbols may refer to more fundamental objects associated with the same operators, so we will occasionally use different notations to distinguish the contexts.

Definition 2.3. There is an $\mathcal{O}$ linear endomorphism (named Verschiebung) τ of $W_{\mathcal{O}}$ defined by

$$\tau(a_0, a_1, \ldots) = (0, a_0, a_1, \ldots),$$

and an endormorphism of $\mathcal{O}$-algebras (named Frobenius) σ characterized by $w_n(\sigma x) = w_{n+1}(x)$ We define Teichmüller lift by $[a] = (a, 0, 0, \ldots)$ for $a \in B$.

If $\pi B = 0$, then

$$\sigma(a_0, a_1, \ldots) = (a_0^q, a_1^q, \ldots).$$

That is why we call it Frobenius.

For the Teichmüller lift, we have

$$[ab] = [a] \cdot [b], \qquad \sigma([a]) = [a^q]$$

for any $a, b \in B$.

The description of the sum $[a] + [b]$ is less straightforward. There exist polynomials

$$P_n \in \mathcal{O}[X, Y]$$

such that

$$[a] + [b] = (a + b, P_1(a, b), P_2(a, b), \ldots).$$

The polynomials P_n can be described recursively using the ghost components of $W_{\mathcal{O}}$. More explicitly, set $P_0(X,Y) = X + Y$. For every $n \geq 1$, the identity

$$\sum_{r=0}^{n} \pi^r P_r(X,Y)^{q^{n-r}} = X^{q^n} + Y^{q^n}$$

gives

$$P_n(X,Y) = \frac{1}{\pi^n}\big(X^{q^n} + Y^{q^n} - (X+Y)^{q^n} - \sum_{r=1}^{n-1} \pi^r P_r(X,Y)^{q^{n-r}}\big).$$

In particular, $X \cdot Y$ divide P_n for all $n \geq 1$. Thus, we have the following corollary.

Corollary 2.4. *If $a \cdot b = 0 \in B$, then $[a] + [b] = [a+b]$ in $W_{\mathcal{O}}(B)$.*

Remark 2.5. We could give them a new characterization if we first establish the Cartier ring.

2.2 The Cartier ring $E_{\mathcal{O}}(B)$

Let B be an $\mathcal{O}$-algebra as usual. Consider a noncommutative $\mathcal{O}$-algebra $W_{\mathcal{O}}(B)[F,V]$ where F and V satisfy the relations:

$$Fx = \sigma(x)F, \qquad xV = V\sigma(x), \qquad VxF = \tau(x), \qquad FV = \pi$$

for $x \in W_{\mathcal{O}}(B)$.

Definition 2.6. The Carier ring $E_{\mathcal{O}}(B)$ is the completion of the above algebra for the topology defined by the right ideals defined by the V^n.

Each element in $E_{\mathcal{O}}(B)$ has a unique convergent expansion

$$\sum_{m,n\geq 0} V^m[a_{m,n}]F^n, \qquad a_{m,n} \in B,$$

subject to the condition: for each m, one has $a_{m,n} = 0$ for n large enough.

We can characterize $W_{\mathcal{O}}(B)$ by $E_{\mathcal{O}}(B)$.

Proposition 2.7. *The map $W_{\mathcal{O}}(B) \to E_{\mathcal{O}}(B)$ given by*

$$(a_0, a_1, \cdots) \mapsto \sum_{n\geq 0} V^n[a_n]F^n$$

is an injective homomorphism of $\mathcal{O}$-algebras.

Then we will consider a special unit in $W_{\mathcal{O}}()B$. Since $w_0(\pi - [\pi]) = 0$, there exists an element $\varepsilon \in W_{\mathcal{O}}(B)$ such that $\pi - [\pi] = V\varepsilon F = \tau(\varepsilon)$. Observe that $w_1(\pi - [\pi]) = \pi(1 - \pi^{q-1})$. Hence, $w_0(\varepsilon) = 1 - \pi^{q-1}$ is a unit in B which implies that ε is a unit.

2.3 Cartier theory for formal $\mathcal{O}$-modules

Definition 2.8. A Cartier $\mathcal{O}$-module over B is a left $E_{\mathcal{O}}(B)$-module M such that

1. M/VM is a free B-module of finite rank;

2. V is injective on M;

3. M is separated and complete for the V-adic topology.

Theorem 2.9 (Cartier equivalence)**.** *The category of formal $\mathcal{O}$-modules over B are equivalent to the category of Cartier $\mathcal{O}$-modules over B. If M_X is the Cartier module corresponding to the formal $\mathcal{O}$ module X, then $M_X/VM_X \cong \mathrm{Lie}(X)$. If B' is a B-algebra, the Cartier module of the formal $\mathcal{O}$-module $X_{B'}$ obtained by base-change is*

$$M' = E_{\mathcal{O}}(B')\widehat{\otimes}_{E_{\mathcal{O}}(B)}M.$$

Definition 2.10. Let M be a Cartier $\mathcal{O}$-module over B. We say that elements $\gamma_1, \ldots, \gamma_d$ of M form a V-basis of M if their images $\bar{\gamma}_1, \ldots, \bar{\gamma}_d$ mod V form a basis for the free B-module M/VM.

Theorem 2.11 (structure theorem)**.** *Every element of M can be written in a unique way as*

$$\sum_{m\geq 0}\sum_{i=1}^{d} V^m[c_{m,i}]\gamma_i$$

with $c_{m,i} \in B$.

In particular, the choice of the γ_i's determines a family $c_{m,i,j}$ $(m \in \mathbb{N};\ i,j \in \{1,\ldots,d\})$ of elements of B such that

$$F(\gamma_j) = \sum_{m\geq 0}\sum_{i=1}^{d} V^m[c_{m,i,j}]\gamma_i, \qquad j = 1,\ldots,d.$$

Conversely, given a family $c_{m,i,j}$ of elements of $W_{\mathcal{O}}(B)$, there exists a Cartier $\mathcal{O}$-module M, unique up to isomorphism, and a V-basis $\gamma_1, \ldots, \gamma_d \in M$ satisfying the above relation.

2.4 Cartier theory for formal $\mathcal{O}_D$-modules

We give the Cartier ring a $\mathbb{Z}/2\mathbb{Z}$-grading defined by $\deg F = \deg V = 1$ and $\deg[b] = 0$ for all $b \in B$. Note that the subring $W_{\mathcal{O}}(B)$ of $E_{\mathcal{O}}(B)$ is contained in the degree 0 component.

Definition 2.12. For an $\mathcal{O}'$-algebra B, a graded Cartier $\mathcal{O}[\Pi]$-module is a Cartier $\mathcal{O}$-module M with a $\mathbb{Z}/2\mathbb{Z}$-grading $M = M_0 \oplus M_1$ and a degree-1 operator Π satisfying $\Pi^2 = \pi$. We say that M is special if M_0/VM_1 and M_1/VM_0 are free of rank 1 over B.

Theorem 2.13 (Specific Cartier equivalence)**.** *The category of formal $\mathcal{O}'[\Pi]$-modules over B are equivalent to the category of graded Cartier $\mathcal{O}[\Pi]$-modules over B. Special objects correspond to special objects.*

Theorem 2.14 (special structure theorem)**.** *Let M be a special graded Cartier $\mathcal{O}[\Pi]$-module over B. If $\bar{y}_i$ is a basis of M_i/VM_{i+1} and y_i is a lift, then (y_0, y_1) is a homogeneous V-basis. Every element can be written in a unique way as*

$$x = \sum_{m\geq 0}\big(V^m[a_{m,0}]y_0 + V^m[a_{m,1}]y_1\big).$$

with $a_{m,i} \in B$.

In particular, the choice of a homogeneous V-basis (γ_0, γ_1) determines elements $a_{m,i}$ $(m \in \mathbb{N},\ i = 0, 1)$ of B such that

$$\Pi\gamma_0 = [a_{0,0}]\gamma_1 + \sum_{m>0} V^m[a_{m,0}]\gamma_{\overline{m+1}},$$
$$\Pi\gamma_1 = [a_{0,1}]\gamma_0 + \sum_{m>0} V^m[a_{m,1}]\gamma_{\overline{m}},$$

such that $a_{0,0}a_{0,1} = \pi$. The converse is also true.

We will explain the main ideas behind the proofs of the above two theorems.

Sketch of the proof of Theorem 2.14. We just need to apply the first part of Theorem 2.11 to M_0 and M_1 respectively, and note that $\Pi^2 = \pi$ and $F = \varepsilon^{-1}V^{-1}(\pi - [\pi])$[4] which means that F is determined by Π; then we apply the second part of Theorem 2.11. □

Sketch of the proof of Theorem 2.13. We should first notice that $\mathcal{O}' = \mathcal{O}[\zeta]$ where ζ is a primitive $q^2 - 1$-th root of 1. Since $[\zeta] \in W_{\mathcal{O}}(B)$ is also a primitive $q^2 - 1$-th root of 1, we find that $W_{\mathcal{O}}(B)$ is actually an $\mathcal{O}'$-algebra. Let X be a formal $\mathcal{O}'[\Pi]$-module and M is its corresponding Cartier $\mathcal{O}$-module by Theorem 2.9, then M is a $W_{\mathcal{O}}(B)$-module and hence a natural $\mathcal{O}'$-module. Since X admits an $\mathcal{O}_D$-action, the Cartier module M also admits a Π-action and an $\mathcal{O}'$-action (extending its $\mathcal{O}$-module structure), and they commute with the action of $E_{\mathcal{O}}(B)$. Thus, by Proposition 1.18, M has a $\mathbb{Z}/2\mathbb{Z}$-grading whence Π, V, F become operators of degree 1 and $[b]$ of degree 0 for $b \in B$.

Conversely, giving the $\mathbb{Z}/2\mathbb{Z}$-grading structure on M is equivalent to give an $\mathcal{O}'$-action ($[\zeta]$-action) on M compatible the original $\mathcal{O}$-action. Giving an action of $\mathcal{O}_D$ is equivalent to giving an action of Π additionally. Then we apply Theorem 2.9. □

Finally, let us consider the problem of base change, which turns out to be just a scalar extension.

Proposition 2.15. *Let B' be a B-algebra and*

$$M' = E_{\mathcal{O}}(B')\widehat{\otimes}_{E_{\mathcal{O}}(B)}M$$

the Cartier module over B' obtained from M by change of base. Then M' is a graded Cartier $\mathcal{O}[\Pi]$-module over B'; the image (γ'_0, γ'_1) in M' of (γ_0, γ_1) is a homogeneous V-basis of M' satisfying the relations

$$\Pi\gamma'_i = [a'_{0,i}]\gamma'_{i+1} + \sum_{m>0} V^m[a'_{m,i}]\gamma'_{m+1+i}, \qquad i = 0, 1,$$

where the $a'_{m,i}$ are the images of the elements $a_{m,i}$ of B inside B'.

Corollary 2.16. *Let B_1 and B_2 be B-algebras. There is a canonical isomorphism*

$$M_{B_1 \times B_2} \cong M_{B_1} \times M_{B_2}.$$

Proof. Define a morphism $\Phi\colon M_{B_1\times B_2} \to M_{B_1} \times M_{B_2}$ on homogeneous generators by

$$\Phi\big(V^m[(a, b)]\gamma_i\big) = \big(V^m[a]\gamma_i, V^m[b]\gamma_i\big).$$

[4]Here V^{-1} is well-defined since V is injective.

Conversely, define $\Psi\colon M_{B_1} \times M_{B_2} \to M_{B_1 \times B_2}$ by

$$\Psi\big(V^m[a]\gamma_i, 0\big) = V^m[(a,0)]\gamma_i \text{ and } \Psi\big(0, V^m[b]\gamma_i\big) = V^m[(0,b)]\gamma_i,$$

and extend Ψ additively.

It remains to verify that Φ and Ψ are mutually inverse. The identity $\Phi \circ \Psi = \mathrm{id}$ is immediate. For the other composition, it suffices to show that $[(a,b)] = [(a,0)] + [(0,b)]$. This follows from Lemma 2.4, since $(a,0)(0,b) = 0$ in $B_1 \times B_2$. Therefore, $\Psi \circ \Phi = \mathrm{id}$, and the result follows. $\square$

3 Isogenies and height

Definition 3.1. For formal groups X_1, X_2 of the same finite dimension, a homomorphism $\varphi : X_1 \to X_2$ is an *isogeny* if its kernel is representable. We say X_1 and X_2 are isogenous if there exists an isogeny between them.

Over an algebraically closed field L of characteristic p, the kernel of an isogeny φ must be representable by

$$R := L[X_1, \ldots, X_n]/(X_1^{q^{h_1}}, \ldots, X_n^{q^{h_n}})$$

for some $n \in \mathbb{N}$ and $h_i \in \mathbb{N}_+$.

In this case, we define the height of φ to be $\log_q \mathrm{rk}_L(R) = h_1 + \cdots + h_n$.

Definition 3.2. For a general base B and a geometric point $x : \operatorname{Spec} L \to \operatorname{Spec} B$, define

$$\mathrm{ht}(\varphi)(x) = \mathrm{ht}(\varphi_x).$$

The formation of the kernel commutes with base change and the dimension of a vector space is stable under base change, so this number depends only on the underlying point $x \in \operatorname{Spec} B$. If the resulting function is constant, that constant is the height of φ.

Proposition 3.3. *If $\varphi : X_1 \to X_2$ and $\psi : X_2 \to X_3$ are morphisms between formal groups, then $\psi\varphi$ is an isogeny if and only if both morphisms are isogenies. In this case,*

$$\mathrm{ht}(\psi\varphi) = \mathrm{ht}(\varphi) + \mathrm{ht}(\psi).$$

Proposition 3.4. *Let $\varphi : X_1 \to X_2$ be an isogeny. Then X_2 is the cokernel of the inclusion $\ker \phi \hookrightarrow X_1$.*

Definition 3.5. A formal $\mathcal{O}$-module X is π-divisible if multiplication by π is an isogeny, and then we define the height of X by $\mathrm{ht}(X) = \mathrm{ht}(\pi)$.[5]

Proposition 3.6. *Let X be a formal $\mathcal{O}$-module over an algebraically closed field L, and let M_X be its Cartier module. Then X is π-divisible if and only if M_X is a finite free $W_{\mathcal{O}}(L)$-module. In this case, we have*

$$\mathrm{ht}(X) = \mathrm{rk}_{W_{\mathcal{O}}(L)} M_X.$$

Fix a special, π-divisible formal $\mathcal{O}_D$-module Φ of height-4 over $\bar{k}$.

Proposition 3.7. *Let X be a special π-divisible formal $\mathcal{O}_D$-module of height-4 over an $\bar{k}$-algebra B. Then*

$$\mathrm{Hom}_{\mathcal{O}_D}(\Phi_B, X)$$

is torsion-free as an $\mathcal{O}$-module.

[5] Here π denotes the endomorphism on X.

Definition 3.8. When B is a $\bar{k}$-algebra, a quasi-isogeny

$$\rho : \Phi_B \dashrightarrow X$$

is an element of

$$\mathrm{Hom}_{\mathcal{O}_D}(\Phi_B, X) \otimes_{\mathcal{O}} K$$

such that $\pi^n \rho$ is an isogeny for all sufficiently large n. Its height is defined by

$$\mathrm{ht}(\rho) = \mathrm{ht}(\pi^n \rho) - \mathrm{ht}(\pi^n) = \mathrm{ht}(\pi^n \rho) - 4n.$$

Remark 3.9. In the category of π-divisible $\mathcal{O}$-modules over some k-algebra B, isogenous is an equivalent relation by Proposition 3.3 and the following proposition.

Proposition 3.10. *If $\varphi : X_1 \to X_2$ is an isogeny between two π-divisible formal $\mathcal{O}$-modules over a k-algebra B, then there exists a morphism $\psi : X_2 \to X_1$ such that $\psi \circ \varphi = \pi^n$ for some $n \in \mathbb{Z}$.*

4 Drinfeld's moduli functor

Let $\widehat{\mathcal{O}^{\mathrm{nr}}}$ be the completion of the maximal unramified extension of $\mathcal{O}$.

Definition 4.1. For a $\widehat{\mathcal{O}^{\mathrm{nr}}}$-algebra B on which π is nilpotent, put

$$G(B) = \left\{ (X, \rho) \;\middle|\; \begin{array}{l} X \text{ is a special, } \pi\text{-divisible formal } O_D\text{-module of height 4 over } B, \\ \rho : \Phi_{B/\pi B} \dashrightarrow X_{B/\pi B} \text{ is a quasi-isogeny of height 0} \end{array} \right\} / \cong .$$

For convenience, we denote the category of $\widehat{\mathcal{O}^{\mathrm{nr}}}$-algebras on which π is nilpotent by $\overline{\mathrm{Nilp}}$.

Let $\widehat{\Omega}$ be Drinfeld's formal p-adic upper half-plane and set

$$\widehat{\Omega}^{\mathrm{nr}} = \widehat{\Omega} \widehat{\otimes}_{\mathcal{O}} \widehat{\mathcal{O}^{\mathrm{nr}}}.$$

Theorem 4.2 (Drinfeld)**.** *The functor G is represented by $\widehat{\Omega}^{\mathrm{nr}}$.*

The proof has eight stages:

Step 1. Identify $\widehat{\Omega}$ with a special functor $F : \mathrm{Nilp} \to \mathbf{Set}$.[6]

Step 2. Construct the "linear algebra" objects η_M, T_M and u_M from the Cartier module M corresponding to a special formal $\mathcal{O}_D$-module over an $\mathcal{O}'$-algebra B. Here η_M is an $\mathcal{O}$-module, T_M is a B-module and u_M is an $\mathcal{O}$-homomorphism from η_M to T_M.

Step 3. Study the propoties of these objects such as $\mathbb{Z}/2\mathbb{Z}$-grading, compatible Π-actions, and so on.

Step 4. Prove the coarse, geometric-point version of Drinfeld's theorem over every algebraically closed $\mathcal{O}$-field L. It means that for any such L, we can define a bejection $\xi(L) : G(L) \to H(L)$ where H is the restriction of F to Nilp.

Step 5. Construct sheaves corresponding to η_M and T_M, equip them with their π-adic filtrations, and construct a morphism r named rigitification. Then construct a natural transformation $\xi : G \to H$.

[6]The categoty Nilp consisting of $\mathcal{O}$-algebras on which π is nilpotent.

Step 6. Reduce the assertion to the special fibre, i.e. restrict functors to the category of $\bar{k}$-algebras.

Step 7. Prove that we have bijection

$$t_G(x) \xrightarrow{\sim} t_H(\xi(x))$$

for every $x \in G(\bar{k})$.

Step 8. Find a cover of G, write $G = \bigcup_{n,m} G_{n,m}$. Prove that every $G_{n,m}$ is represented by a projective scheme. Then complete the proof by checking the gluing data on both sides.

The mathematical prerequisites also separate by step. Steps 1–2 are mainly linear algebra together with the relations in the Cartier ring. Step 3 uses the theory of π-divisible groups over an algebraically closed field of characteristic p. Step 4 uses the étale topology and constructible sheaves. Step 5 needs the deformation theory for formal groups. Step 6 needs the deformation theory and some direct linear algebra calculations. Step 7 uses isogenies, finite locally free quotients, Grassmannians, and elementary algebraic geometry.

For the definitions and properties of the p-adic upper half plane $\widehat{\Omega}$, the functor F, and Step 1, we refer to my previous reading report on the first part of this paper.

5 Construction of η_M, T_M and u_M

Throughout this section, B denotes an $\mathcal{O}'$-algebra.

5.1 Some basic constructions

Definition 5.1. Define

$$E'_{\mathcal{O}}(B) = W_{\mathcal{O}}(B)[V, \Pi]$$

to be the ring generated over $W_{\mathcal{O}}(B)$ by V and Π, subject to the relations

$$\Pi V = V\Pi, \qquad \Pi x = x\Pi, \qquad xV = V\sigma(x), \qquad \Pi^2 = \pi$$

for every $x \in W_{\mathcal{O}}(B)$.

Thus, an $E'_{\mathcal{O}}(B)$-module is equivalently a $W_{\mathcal{O}}(B)$-module equipped with operators V and Π satisfying the preceding relations. In particular, V is semilinear and Π is $W_{\mathcal{O}}(B)$-linear. We endow $E'_{\mathcal{O}}(B)$ with a $\mathbb{Z}/2\mathbb{Z}$-grading by declaring

$$\deg(V) = \deg(\Pi) = 1, \qquad \deg(x) = 0 \quad \text{for } x \in W_{\mathcal{O}}(B).$$

Unless otherwise specified, M will denote a $W_{\mathcal{O}}(B)$-module. Let M^σ be the scalar twist of M along σ^{-1}; explicitly, M^σ has the same underlying abelian group as M, and its scalar multiplication is given by

$$x \cdot_\sigma m = \sigma(x)m.$$

If M is an $E'_{\mathcal{O}}(B)$-module, then so is M^σ, and the operator V defines an $E'_{\mathcal{O}}(B)$-linear homomorphism $V\colon M^\sigma \longrightarrow M$ which is homogeneous of degree 1.

Definition 5.2. For an $E_{\mathcal{O}}(B)$-module M, set

$$N(M) = \operatorname{coker}\Big(M^\sigma \xrightarrow{\alpha_M} M \oplus M^\sigma\Big), \qquad \text{where } \alpha_M(m) = (Vm, -\Pi m).$$

We write

$$\beta_M \colon M \oplus M^\sigma \longrightarrow N(M)$$

for the quotient map and abbreviate $\beta_M(m, m')$ to $\overline{(m, m')}$. Then $N(M)$ is naturally an $E'_{\mathcal{O}}(B)$-module.

If $M = M_0 \oplus M_1$ is graded, then $N(M)$ inherits a grading

$$N(M) = N(M)_0 \oplus N(M)_1, \qquad N(M)_i = \beta_M(M_i \oplus M_i^\sigma).$$

To verify that this sum is direct, suppose that

$$\beta_M(m_0, m_0') = \beta_M(m_1, m_1'), \qquad (m_i, m_i') \in M_i \oplus M_i^\sigma.$$

Then there exists $n = n_0 + n_1 \in M^\sigma$ such that

$$(m_0 - m_1, m_0' - m_1') = (Vn, -\Pi n).$$

By comparing the homogeneous components, we obtain

$$(m_0, m_0') = (Vn_1, -\Pi n_1), \qquad (m_1, m_1') = (-Vn_0, \Pi n_0).$$

Consequently, both classes vanish, and hence $N(M)_0 \cap N(M)_1 = 0$.

Proposition 5.3. *The functor*

$$N \colon \operatorname{Mod}_{E'_{\mathcal{O}}(B)} \longrightarrow \operatorname{Mod}_{E'_{\mathcal{O}}(B)}$$

is right exact. Moreover, it carries an exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

to an exact sequence

$$0 \longrightarrow N(M') \longrightarrow N(M) \longrightarrow N(M'') \longrightarrow 0$$

provided that $V \colon M'' \to M''$ is injective.

Proof. Right exactness follows directly from the cokernel construction. For the assertion on short exact sequences, apply the snake lemma to the diagram defining $N(M')$, $N(M)$, and $N(M'')$. The only possible obstruction to injectivity on the left is the kernel of V on M'', which vanishes by assumption. □

We record several natural morphisms of $E'_{\mathcal{O}}(B)$-modules:

1. The map
$$\iota_1 \colon M \longrightarrow N(M), \qquad m \longmapsto \overline{(m, 0)},$$
is injective whenever $\Pi \colon M \to M$ is injective.

2. The map
$$\iota_2 \colon M^\sigma \longrightarrow N(M), \qquad m \longmapsto \overline{(0, m)},$$
is injective whenever $V \colon M \to M$ is injective.

3. There is a natural map

$$P\colon N(M) \longrightarrow M/VM, \qquad \overline{(m,m')} \longmapsto \overline{m} \pmod{VM}.$$

4. There is a natural map

$$\lambda_M\colon N(M) \longrightarrow M, \qquad \overline{(m,m')} \longmapsto \Pi m + Vm'.$$

The maps ι_1, ι_2, and P have degree 0, whereas λ_M has degree 1. Moreover, λ_M is $E'_{\mathcal{O}}(B)$-linear. For example,

$$\lambda_M\big(x \cdot \overline{(m,m')}\big) = \Pi(xm) + V(\sigma(x)m') = x(\Pi m + Vm') = x\lambda_M\big(\overline{(m,m')}\big).$$

Lemma 5.4. *Let $B \twoheadrightarrow B'$ be a surjective homomorphism of $\mathcal{O}$-algebras with kernel I, and let M be a Cartier $\mathcal{O}$-module over B. Set*

$$M' = E'_{\mathcal{O}}(B') \mathbin{\widehat{\otimes}}_{E'_{\mathcal{O}}(B)} M.$$

Choose a V-basis $\{y_i\}_i$ of M, and define

$$M_I = \left\{ \sum_{m,i} V^m[a_{m,i}]y_i \;\middle|\; a_{m,i} \in I \right\}.$$

Then the sequence

$$0 \longrightarrow M_I \longrightarrow M \longrightarrow M' \longrightarrow 0$$

is exact. Consequently,

$$0 \longrightarrow N(M_I) \longrightarrow N(M) \longrightarrow N(M') \longrightarrow 0$$

is exact.

Proof. Recall the structure theorem for Cartier $\mathcal{O}$-modules. The images of the y_i form a B-basis of M/VM, and every element of M admits a unique expansion

$$\sum_{m,i} V^m[a_{m,i}]y_i, \qquad a_{m,i} \in B.$$

Similarly, every element of M' admits a unique expansion of the same form with coefficients in B'. The kernel of $M \to M'$ therefore consists precisely of those expansions whose coefficients lie in I, namely M_I. This proves the first exact sequence. Since M' is again a Cartier module, V is injective on M', and the second exact sequence follows from Proposition 5.3. □

5.2 The map L_M

Proposition 5.5. *Let M be a special graded Cartier $\mathcal{O}[\Pi]$-module. There exists an additive homomorphism*

$$L_M\colon M \longrightarrow N(M)$$

that is compatible with base change and satisfies

$$F = \lambda_M \circ L_M.$$

More precisely, for $a \in W_{\mathcal{O}}(B)$ and $x \in M$, one has

$$L_M(ax) = \sigma(a)L_M(x), \qquad L_M(Vx) = \overline{(\Pi x, 0)}, \qquad L_M \circ \Pi = \Pi \circ L_M.$$

Furthermore, L_M is homogeneous of degree 0.

Proof. We first assume that B is π-torsion-free. In this case, Proposition 5.12, proved in the appendix, shows that λ_M is injective. It is therefore enough to prove

$$F(M) \subseteq \lambda_M(N(M)) = \Pi M + VM.$$

Once this inclusion is established, L_M is uniquely defined by the identity $F = \lambda_M \circ L_M$.

Let (y_0, y_1) be a homogeneous V-basis, so that

$$M_0/VM_1 = B\overline{y_0}, \qquad M_1/VM_0 = B\overline{y_1}.$$

By the uniqueness of Cartier expansions,

$$M = \{[a_0]y_0 + [a_1]y_1 \;\mid\; a_0, a_1 \in B\} + VM.$$

Since

$$F(VM) = \pi M = \Pi^2 M \subseteq \Pi M + VM,$$

it remains to prove that $Fy_i \in \Pi M + VM$ for $i = 0, 1$.

Recall that

$$\pi - [\pi] = V\varepsilon F$$

for some unit ε in the Cartier ring, and that V is injective. It therefore suffices to show

$$(\pi - [\pi])y_i \in V(\Pi M + VM).$$

Write

$$\Pi y_0 = [a_{00}]y_1 + Vx_0, \qquad \Pi y_1 = [a_{01}]y_0 + Vx_1,$$

where $a_{00}, a_{01} \in B$ and $x_i \in M_i$. We know that$a_{00}a_{01} = \pi$ by Theorem 2.9. Consequently,

$$\begin{aligned}(\Pi^2 - [\pi])y_0 &= \Pi\big([a_{00}]y_1 + Vx_0\big) - [\pi]y_0 \\ &= [a_{00}][a_{01}]y_0 + [a_{00}]Vx_1 + \Pi Vx_0 - [\pi]y_0 \\ &= [a_{00}]Vx_1 + \Pi Vx_0 \\ &= V\big([a_{00}^q]x_1 + \Pi x_0\big) \\ &\in V\big([a_{00}]M_1 + \Pi M_0\big).\end{aligned}$$

On the other hand,

$$[a_{00}]y_1 = \Pi y_0 - Vx_0 \in \Pi M + VM.$$

It then implies that $[a_{00}]M_1 \subseteq \Pi M + VM$, and hence $[a_{00}^q]M_1 \subseteq \Pi M + VM$. Thus

$$(\pi - [\pi])y_0 \in V(\Pi M + VM).$$

The argument for y_1 is identical.

For a general $\mathcal{O}'$-algebra B, choose the canonical surjection

$$\mathcal{O}[X_b \mid b \in B] \twoheadrightarrow B, \qquad X_b \longmapsto b.$$

The source is π-torsion-free. Applying the construction there and then using Lemma 5.4 yields L_M over B; the same lemma also shows that the result is independent of the chosen presentation. Finally, all the displayed identities follow after applying λ_M to both sides and using the injectivity argument over a torsion-free presentation. This also proves compatibility with base change. We omit the details. □

5.3 The morphism ϕ_M

Definition 5.6. Define an endomorphism

$$\begin{aligned} \phi_M \colon N(M) &\longrightarrow N(M) \\ \big(\overline{(x,x')}\big) &\mapsto L_M(x) + \overline{(x',0)}. \end{aligned}$$

This map is well defined, because

$$\phi_M\big(\overline{(Vx,-\Pi x)}\big) = L_M(Vx) - \overline{(\Pi x,0)} = 0.$$

It is additive, homogeneous of degree 0, and satisfies

$$\phi_M \circ \Pi = \Pi \circ \phi_M, \qquad \phi_M(ay) = \sigma(a)\phi_M(y).$$

In particular, ϕ_M is $\mathcal{O}[\Pi]$-linear.

Definition 5.7. Let M be a special graded Cartier $\mathcal{O}[\Pi]$-module. Define

$$\eta_M = N(M)^{\phi_M} = \{y \in N(M) \mid \phi_M(y) = y\}, \qquad T_M = M/VM,$$

and let $u_M \colon \eta_M \longrightarrow T_M$ be the composite

$$\eta_M \hookrightarrow N(M) \xrightarrow{P} M/VM.$$

Here η_M is naturally an $\mathcal{O}[\Pi]$-module, whereas T_M is naturally a $B[\Pi]$-module. The morphism u_M is $\mathcal{O}[\Pi]$-linear and is compatible with base change.

Proposition 5.8. *Let*

$$0 \longrightarrow I \longrightarrow B \longrightarrow B' \longrightarrow 0$$

be an exact sequence of $\mathcal{O}$-algebras. Assume that $I^N = 0$ and $\pi^N I = 0$ for some sufficiently large integer N. If

$$M' = E_{\mathcal{O}}(B') \mathbin{\widehat{\otimes}}_{E_{\mathcal{O}}(B)} M,$$

then the natural map $\eta_M \longrightarrow \eta_{M'}$ is a bijection.

Proof. By induction on the nilpotence exponents, it is enough to treat the case

$$I^2 = 0, \qquad \pi I = 0.$$

By Lemma 5.4, there is an exact sequence

$$0 \longrightarrow M_I \longrightarrow M \longrightarrow M' \longrightarrow 0.$$

For $x \in M_I$, write

$$x = [a_0]y_0 + [a_1]y_1 + Vx', \qquad a_0, a_1 \in I, \qquad x' \in M_I.$$

Since $I^2 = 0$, and hence $a_i^q = 0$, we obtain

$$\begin{aligned} \phi_M\big(\overline{(x,0)}\big) &= [a_0^q]\phi_M\big(\overline{(y_0,0)}\big) + [a_1^q]\phi_M\big(\overline{(y_1,0)}\big) + L_M(Vx') \\ &= \overline{(\Pi x',0)}. \end{aligned}$$

It follows that

$$\phi_M^2\big(\overline{(x,0)}\big) = \overline{(\Pi^2 x'',0)} = \overline{(\pi x'',0)} = 0$$

for some $x'' \in M_I$. For the last equality, one uses, for $a \in I$, the identity

$$\pi[a] = [\pi a] + V\varepsilon F[a] = [\pi a] + V\varepsilon[a^q]F = 0,$$

because $\pi I = 0$ and $I^2 = 0$. Moreover, $\phi_M\big(\overline{(0,x)}\big) = \overline{(x,0)}$. Therefore, ϕ_M^3 vanishes on $N(M_I)$.

Proposition 5.3 gives a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N(M_I) & \longrightarrow & N(M) & \longrightarrow & N(M') & \longrightarrow & 0 \\ & & {\scriptstyle 1-\phi_M}\downarrow & & {\scriptstyle 1-\phi_M}\downarrow & & {\scriptstyle 1-\phi_{M'}}\downarrow & & \\ 0 & \longrightarrow & N(M_I) & \longrightarrow & N(M) & \longrightarrow & N(M') & \longrightarrow & 0. \end{array}$$

Because $\phi_M^3 = 0$ on $N(M_I)$, the endomorphism $1-\phi_M$ is invertible there, with inverse $1+\phi_M+\phi_M^2$. The snake lemma therefore yields the required bijection $\eta_M \xrightarrow{\sim} \eta_{M'}$. □

Remark 5.9. From the proof, we can deduce that, under the same assumption, $\phi_M|_{N(M_I)}$ is nilpotent.

5.4 Appendix: Proof of the injectivity assertion

Lemma 5.10. *If $\pi \in B^\times$, equivalently if B is a K-algebra, then λ_M is an isomorphism.*

Proof. In this case, the ghost map w is bijective. Hence $\pi \in W_{\mathcal{O}}(B)^\times$, and therefore Π is invertible because $\Pi^2 = \pi$. It follows that

$$\iota_1 \colon M \longrightarrow N(M), \qquad m \longmapsto \overline{(m,0)},$$

is an isomorphism: injectivity follows from the injectivity of Π, and surjectivity follows from the relation:

$$\overline{(0,m')} = \overline{(V\Pi^{-1}m',0)}.$$

Since $\lambda_M \circ \iota_1 = \Pi$, the map λ_M is an isomorphism as well. □

Lemma 5.11. *Assume that π is a non-zero-divisor in B, and let M be a Cartier $\mathcal{O}$-module over B. Put*

$$B_K = B \otimes_{\mathcal{O}} K, \qquad M_K = M \mathbin{\widehat{\otimes}}_{E_{\mathcal{O}}(B)} E_{\mathcal{O}}(B_K).$$

Then the natural map $M \to M_K$ is injective, and V acts injectively on M_K/M.

Proof. Choose a V-basis $\{y_i\}_i$ of M. The unique Cartier expansions give

$$M = \left\{ \sum_{i,m} V^m[a_{m,i}]y_i \;\middle|\; a_{m,i} \in B \right\} \text{ and } M_K = \left\{ \sum_{i,m} V^m[a'_{m,i}]y_{i,K} \;\middle|\; a'_{m,i} \in B_K \right\}.$$

Under these descriptions, the natural map $M \to M_K$ is induced coefficientwise by

$$B \longrightarrow B_K = B[\pi^{-1}],$$

which is injective because π is a non-zero-divisor in B. Hence $M \to M_K$ is injective.

Now let $x' \in M_K$ and suppose that $Vx' \in M$. By comparing the unique Cartier expansions of Vx' in M and M_K, one finds that every coefficient of x' belongs to B. Thus $x' \in M$, proving that V is injective on M_K/M. □

Lemma 5.12. *Assume that π is a non-zero-divisor in B, and let M be a special graded Cartier $\mathcal{O}[\Pi]$-module over B. Then λ_M is injective.*

Proof. By Lemma 5.11, the sequence

$$0 \longrightarrow M \longrightarrow M_K \longrightarrow M_K/M \longrightarrow 0$$

is exact, and V is injective on M_K/M. Proposition 5.3 therefore gives an exact sequence

$$0 \longrightarrow N(M) \longrightarrow N(M_K) \longrightarrow N(M_K/M) \longrightarrow 0.$$

Consider the commutative square

$$\begin{array}{ccc} N(M) & \xrightarrow{\lambda_M} & M \\ \downarrow & & \downarrow \\ N(M_K) & \xrightarrow[\lambda_{M_K}]{} & M_K. \end{array}$$

The left vertical arrow is injective by the preceding exact sequence, the right vertical arrow is injective by Lemma 5.11, and the lower horizontal arrow is an isomorphism by Lemma 5.10. It follows that λ_M is injective. □

6 Some calculations on η_M

Let B be an $\mathcal{O}'$-algebra such that $\pi B = 0$, and let $M = M_0 \oplus M_1$ be a special graded Cartier $\mathcal{O}[\Pi]$-module. All indices in what follows are taken modulo 2. Put $T_i = M_i/VM_{i+1}$.

Definition 6.1. An index $i \in \{0, 1\}$ is called *critical* if the induced map $\Pi\colon T_i \longrightarrow T_{i+1}$ is zero. Equivalently, $\Pi M_i \subseteq VM_i$. In this case, since V is injective, the operator $V^{-1}\Pi$ is hence a well-defined endomorphism on M_i.

Lemma 6.2. *If i is critical, then $L_M(x) = \overline{(V^{-1}\Pi x, 0)}$ for every $x \in M_i$.*

Proof. Choose a homogeneous V-basis $\{y_0, y_1\}$ of M. Suppose that the formula has been proved for $x = y_i$. Every $x \in M_i$ can be written as $x = [a]y_i + Vx'$ for some $a \in B$ and $x' \in M_{i+1}$. Hence

$$\begin{aligned} L_M(x) &= [a^q]L_M(y_i) + L_M(Vx') \\ &= [a^q]\overline{(V^{-1}\Pi y_i, 0)} + \overline{(\Pi x', 0)} \\ &= \overline{(V^{-1}\Pi x, 0)}. \end{aligned}$$

It therefore suffices to prove the formula for y_i. Without loss of generality, assume that $i = 0$.

Consider the surjection

$$\widetilde{B} = \mathcal{O}[X_b \mid b \in B] \longrightarrow B, \qquad X_b \longmapsto b.$$

Choose a lift $\widetilde{M}$ of M to a special graded Cartier $\mathcal{O}[\Pi]$-module over $\widetilde{B}$, together with lifts $\widetilde{y}_i$ of y_i. The construction of L_M is compatible with this base change, so we have a commutative diagram

$$\begin{array}{ccccc} \widetilde{M} & \longrightarrow & M & \longrightarrow & 0 \\ {\scriptstyle L_{\widetilde{M}}}\downarrow & & \downarrow{\scriptstyle L_M} & & \\ N(\widetilde{M}) & \longrightarrow & N(M) & \longrightarrow & 0. \end{array}$$

Write

$$\Pi\widetilde{y}_i = [\widetilde{a}_i]\widetilde{y}_{i+1} + V\widetilde{x}_i.$$

Since 0 is critical, $\widetilde{a}_0$ maps to 0 in B. Consequently,

$$\Pi y_0 = V x_0,$$

where $\widetilde{x}_0$ maps to x_0. Thus, it remains to prove that

$$L_M(y_0) = \overline{(x_0, 0)}.$$

Using $\widetilde{a}_0\widetilde{a}_1 = \pi$, we obtain

$$\begin{aligned}\Pi^2\widetilde{y}_0 &= [\widetilde{a}_0]\big([\widetilde{a}_1]\widetilde{y}_0 + V\widetilde{x}_1\big) + \Pi V\widetilde{x}_0\\ &= [\pi]\widetilde{y}_0 + [\widetilde{a}_0]V\widetilde{x}_1 + \Pi V\widetilde{x}_0.\end{aligned}$$

Recall that $\pi - [\pi] = V\varepsilon F$ for some unit $\varepsilon \in E_{\mathcal{O}}(\widetilde{B})$. It follows that

$$V\varepsilon F\widetilde{y}_0 = (\Pi^2 - [\pi])\widetilde{y}_0 = [\widetilde{a}_0]V\widetilde{x}_1 + \Pi V\widetilde{x}_0.$$

Write

$$\widetilde{x}_1 = [b]\widetilde{y}_1 + V\widetilde{z}_0.$$

Then

$$\begin{aligned}[\widetilde{a}_0^q]\widetilde{x}_1 &= [b][\widetilde{a}_0^q]\widetilde{y}_1 + V[\widetilde{a}_0^q]\widetilde{z}_0\\ &= [b]\big(\Pi\widetilde{y}_0 - V\widetilde{x}_0\big) + V[\widetilde{a}_0^q]\widetilde{z}_0\\ &= \Pi[b]\widetilde{y}_0 + V\big([\widetilde{a}_0^q]\widetilde{z}_0 - [b]\widetilde{x}_0\big) \in \Pi\widetilde{M}_0 + V\widetilde{M}_0.\end{aligned}$$

Therefore, there exist $\widetilde{u}_0, \widetilde{v}_0 \in \widetilde{M}_0$ such that

$$[\widetilde{a}_0^q]\widetilde{x}_1 = \Pi[\widetilde{a}_0]\widetilde{u}_0 + V[\widetilde{a}_0]\widetilde{v}_0.$$

Consequently,

$$[\widetilde{a}_0]V\widetilde{x}_1 = V\Pi[\widetilde{a}_0]\widetilde{u}_0 + V^2[\widetilde{a}_0]\widetilde{v}_0,$$

and hence

$$\varepsilon F\widetilde{y}_0 = \Pi[\widetilde{a}_0]\widetilde{u}_0 + V[\widetilde{a}_0]\widetilde{v}_0 + \Pi\widetilde{x}_0.$$

It follows that

$$\lambda_{\widetilde{M}} L_{\widetilde{M}}(\widetilde{y}_0) = \varepsilon^{-1}\big(\Pi[\widetilde{a}_0]\widetilde{u}_0 + V[\widetilde{a}_0]\widetilde{v}_0 + \Pi\widetilde{x}_0\big).$$

Since $\widetilde{B}$ is π-torsion-free, the map $\lambda_{\widetilde{M}}$ is injective. Therefore,

$$L_{\widetilde{M}}(\widetilde{y}_0) = \overline{(\varepsilon^{-1}\widetilde{x}_0 + \varepsilon^{-1}[\widetilde{a}_0]\widetilde{u}_0,\ \sigma(\varepsilon^{-1})[\widetilde{a}_0]\widetilde{v}_0)}.$$

Under the map $E_{\mathcal{O}}(\widetilde{B}) \to E_{\mathcal{O}}(B)$, one has

$$[\widetilde{a}_0] \longmapsto 0, \qquad \varepsilon \longmapsto 1,$$

because $VF = \pi$ over B. Thus, $L_M(y_0) = \overline{(x_0, 0)}$, as required. □

Lemma 6.3. *If i is critical, then the map*

$$\rho\colon N(M)_i \longrightarrow T_i \oplus M_i^{\sigma}, \qquad \overline{(m, m')} \longmapsto \big(\overline{m}, V^{-1}\Pi m + m'\big),$$

is an isomorphism of $W_{\mathcal{O}}(B)$-modules.

Proof. The map ρ is $W_{\mathcal{O}}(B)$-linear, and its inverse is given by

$$\rho^{-1}\colon T_i \oplus M_i^{\sigma} \longrightarrow N(M)_i, \qquad (\overline{m}, m'') \longmapsto \overline{(m, -V^{-1}\Pi m + m'')}.$$

□

Lemma 6.4. *If i is critical, then we have:*

$$\begin{aligned} \rho\varphi_M\rho^{-1}\colon T_i \oplus M_i^{\sigma} &\longrightarrow T_i \oplus M_i^{\sigma} \\ (\overline{m}, m'') &\longmapsto (\overline{m''}, V^{-1}\Pi m''). \end{aligned}$$

Proof. For the first summand, Lemma 6.2 gives

$$(\overline{m}, 0) \longmapsto \overline{(m, -V^{-1}\Pi m)} \longmapsto L_M(m) + \overline{(-V^{-1}\Pi m, 0)} = 0.$$

For the second summand, we have

$$(0, m'') \longmapsto \overline{(0, m'')} \longmapsto \overline{(m'', 0)} \longmapsto (\overline{m''}, V^{-1}\Pi m'').$$

□

Proposition 6.5. *If i is critical, then*

$$V^{-1}\lambda_M\colon \eta_{M,i} = N(M)_i^{\varphi_M} \xrightarrow{\cong} M_i^{V^{-1}\Pi}$$

is an isomorphism of $\mathcal{O}$-modules.

Proof. We have:

$$\eta_{M,i} = N(M)_i^{\varphi_M} \underset{\cong}{\xrightarrow{\rho}} \rho(N(M)_i)^{\rho\varphi_M\rho^{-1}} \underset{=}{\xrightarrow{\text{Lemma 6.3}}} (T_i \oplus M_i^{\sigma})^{\rho\varphi_M\rho^{-1}} \xrightarrow[\cong]{\text{Lemma 6.4}} M_i^{V^{-1}\Pi}$$

□

We now assume that j is not critical and that

$$\Pi\colon T_j = M_j/VM_{j+1} \xrightarrow{\sim} T_{j+1} = M_{j+1}/VM_j$$

is an isomorphism. This condition is automatic by the fact that j is non-critical when B is a field. Notice that it implies that $j+1$ is critical. Indeed,

$$\begin{aligned} \Pi M_{j+1} &= \Pi(\Pi M_j + VM_j) \\ &= \pi M_j + V\Pi M_j \\ &= VFM_j + V\Pi M_j \\ &\subseteq VM_{j+1}. \end{aligned}$$

Lemma 6.6. *The map*

$$\lambda_M\colon N(M)_j \xrightarrow{\sim} M_{j+1}$$

is an isomorphism of $W_{\mathcal{O}}(B)$-modules.

Proof. The map λ_M is surjective because $M_{j+1} = \Pi M_j + VM_j$ by the assumption. To prove injectivity, suppose that

$$\lambda_M\big(\overline{(m,m')}\big) = \Pi m + Vm' = 0.$$

Then $\Pi m \in VM_j$. Since $\Pi\colon T_j \to T_{j+1}$ is an isomorphism, it follows that $m = Vm''$ for some $m'' \in M_{j+1}$. Hence

$$V(\Pi m'' + m') = 0.$$

Since V is injective, $\Pi m'' + m' = 0$, and therefore

$$\overline{(m,m')} = \overline{(Vm'', -\Pi m'')} = 0.$$

□

Lemma 6.7. *On M_{j+1}, one has $\lambda_M \varphi_M \lambda_M^{-1} = V^{-1}\Pi$.*

Proof. For $\overline{(m,m')} \in N(M)_j$, consider the two composites:

$$\overline{(m,m')} \overset{\varphi_M}{\longmapsto} L_M(m) + \overline{(m',0)} \overset{\lambda_M}{\longmapsto} Fm + \Pi m',$$
$$\overline{(m,m')} \overset{\lambda_M}{\longmapsto} \Pi m + Vm' \overset{V^{-1}\Pi}{\longmapsto} Fm + \Pi m'.$$

Here we have used $VF = FV = \pi = \Pi^2$. Then the assertion folows by the injectivity of λ_M. □

Proposition 6.8. *The map λ_M induces an isomorphism of $\mathcal{O}$-modules*

$$\lambda_M\colon \eta_{M,j} = N(M)_j^{\varphi_M} \xrightarrow{\sim} M_{j+1}^{V^{-1}\Pi}.$$

Moreover, we have the following commutative diagram:

$$\begin{array}{ccc} M_{j+1}^{V^{-1}\Pi} & = & M_{j+1}^{V^{-1}\Pi} \\ {\scriptstyle\lambda_M}\uparrow & \cong & \cong \uparrow{\scriptstyle V^{-1}\lambda_M} \\ \eta_{M,j} & \xrightarrow[\Pi]{} & \eta_{M,j+1} \end{array}$$

Proof. The first assertion follows immediately from Lemmas 6.6 and 6.7. The second assertion follows from the commutative diagram:

$$\begin{array}{ccc} M_{j+1} & \xrightarrow{V^{-1}\Pi} & M_{j+1} \\ {\scriptstyle\lambda_M}\uparrow & & \uparrow{\scriptstyle V^{-1}\lambda_M} \\ N(M)_j & \xrightarrow[\Pi]{} & N(M)_{j+1} \end{array}$$

□

Corollary 6.9. *Under the preceding assumptions, the map*

$$\Pi\colon \eta_{M,j} \xrightarrow{\sim} \eta_{M,j+1}$$

is an isomorphism.

Remark 6.10. When B is a field, we can summarize the above results as follows:

1. if i is critical, then $\eta_{M,i} \cong M_i^{V^{-1}\Pi}$;
2. if j is not critical, then $\Pi\colon \eta_{M,j} \xrightarrow{\sim} \eta_{M,j+1}$.

7 Coase moduli problem: classification over an algebraically closed field

Let L be an algebraically closed field of characteristic p, and set

$$\mathcal{W} = W_{\mathcal{O}}(L), \qquad \mathcal{K} = \mathrm{Frac}(\mathcal{W}).$$

Then $\mathcal{W}$ is a DVR with uniformizer π and residue field L. In this case, the Frobenius σ will be an automorphism, and we have $\mathcal{W}^\sigma = \mathcal{O}$ and $\mathcal{K}^\sigma = K$.

In this section, we further require special formal $\mathcal{O}_D$-modules to be π-divisible.

7.1 Special formal $\mathcal{O}_D$-module of height-4

Proposition 7.1. *If X is a special formal $\mathcal{O}_D$-module, then* $\mathrm{ht}(X) \in 4\mathbb{Z}$.

Proof. Let $M = M_0 \oplus M_1$ be the corresponding graded Cartier $\mathcal{O}[\Pi]$-module. Then M is a finite free $\mathcal{W}$-module. Thus, M_0 and M_1 are finitely generated torsion-free $\mathcal{W}$-modules, and hence both are finite free $\mathcal{W}$-modules. Since

$$V\colon M_i \to M_{i+1}$$

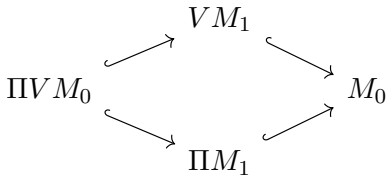
is injective for each $i \in \mathbb{Z}/2\mathbb{Z}$, we have

$$\mathrm{rk}_{\mathcal{W}} M_0 = \mathrm{rk}_{\mathcal{W}} M_1.$$

We denote this common rank by r. Specialness gives

$$[M_0 : VM_1] = [M_1 : VM_0] = 1.$$

The commutative diamond



and the injectivity of V imply

$$[M_0 : VM_1] + [M_1 : \Pi M_0] = [M_0 : \Pi M_1] + [M_1 : VM_0].$$

The two outer terms are both 1, hence

$$[M_0 : \Pi M_1] = [M_1 : \Pi M_0] =: s.$$

Using $\Pi^2 = \pi$,

$$\begin{aligned} r = [M_0 : \pi M_0] &= [M_0 : \Pi M_1] + [\Pi M_1 : \pi M_0] \\ &= [M_0 : \Pi M_1] + [M_1 : \Pi M_0] = 2s. \end{aligned}$$

Therefore

$$\mathrm{ht}(X) = \mathrm{rk}_W M = 2r = 4s \in 4\mathbb{Z}.$$

□

We will study special formal $\mathcal{O}_D$-modules of height-4. We then recall the classification theory of π-divisible formal $\mathcal{O}$-modules in terms of isocrystals.

Definition 7.2. A crystal is a finite free $\mathcal{W}$-module M with a σ^{-1}-linear map $\Phi : M \to M$. An isocrystal is a finite-dimensional $\mathcal{K}$-vector space N with a σ^{-1}-linear automorphism $\Phi : N \to N$.

For a π-divisible formal $\mathcal{O}$-module, (M, V) is a crystal, while $(M \otimes_W \mathcal{K}, V)$ is the associated isocrystal.

Lemma 7.3. *Let M_1, M_2 be Cartier modules corresponding to two π-divisible formal $\mathcal{O}$-modules. Then for any morphism of crystals*

$$\varphi : (M_1, V) \to (M_2, V),$$

it is automatically a morphism of Cartier $\mathcal{O}$-modules.

Proof. It suffices to show that φ commutes with the operator F. It follows immediately since $F = V^{-1}\pi$ in this case. □

Lemma 7.4. *If $\varphi : X_1 \to X_2$ is a morphism between two π-divisible formal $\mathcal{O}$-module, then φ is an isogeny if and only if φ induces an isomorphism between the corresponding isocystals of X_1 and X_2.*

Theorem 7.5. *Let X_1, X_2 be π-divisible formal $\mathcal{O}$-modules. Then X_1 and X_2 are isogenous if and only if their corresponding isocrystals are isomorphic.*

Proof. If X_1 and X_2 are isogenous, then the corresponding isocrystals are clearly isomorphic by Lemma 7.4.

Let M_i be the Cartier $\mathcal{O}$-module corresponding to X_i, $i = 1, 2$.

If there is an isomorphism

$$\varphi : (M_1 \otimes \mathcal{K}, V) \cong (M_2 \otimes \mathcal{K}, V),$$

then we have

$$\pi^n\varphi \colon (M_1, V) \to (M_2, V)$$

is a morphism of crystals for some $n \in \mathbb{N}$. By the previous lemma and Theorem 2.9, we obtain a morphism $\psi \colon X_1 \to X_2$ corresponding to $\pi^n\varphi$. Thus, the morphism ψ is an isogeny by Lemma 7.4. □

We now return to special formal $\mathcal{O}_D$-modules of height-4.

Proposition 7.6. *Any two special formal $\mathcal{O}_D$-modules of height-4 over L are isogenous.*

Proof. Let $M = M_0 \oplus M_1$ be the Cartier module of X. Because M_i/VM_{i-1} is one-dimensional over L, the composite

$$M_i/VM_{i-1} \xrightarrow{\Pi} M_{i+1}/VM_i \xrightarrow{\Pi} M_i/VM_{i-1}$$

is multiplication by $\Pi^2 = \pi = 0$. Hence at least one of the two maps is zero, so at least one index is critical. After renumbering, assume that 0 is critical.

Then the isocrystal $(M \otimes_W \mathcal{K}, V)$ is determined by $(M_0 \otimes_W \mathcal{K}, V\Pi^{-1})$, because $\Pi : M_1 \otimes_W \mathcal{K} \to M_0 \otimes_W \mathcal{K}$ is an isomorphism. The height-4 calculation gives

$$[M_1 : \Pi M_0] = [M_1 : VM_0] = 1.$$

Together with $\Pi M_0 \subseteq VM_0$ at the critical index, this yields $\Pi M_0 = VM_0$. Therefore $V\Pi^{-1}$ preserves M_0 and, after choosing a lattice, the resulting two-dimensional isocrystal is the standard one[7]

$$(\mathcal{K}^2, \Phi), \qquad \Phi(ae_1 + be_2) = \sigma^{-1}(a)e_1 + \sigma^{-1}(b)e_2.$$

Thus every height-4 object has the same isocrystal. By the preceding theorem, any two such formal $\mathcal{O}_D$-modules are isogenous. □

Proposition 7.7. *For a height-4 special formal $\mathcal{O}_D$-module X,*

$$\mathrm{End}_D^0(X) := \mathrm{End}_{\mathcal{O}_D}(X) \otimes_{\mathbb{Z}} \mathbb{Q} \cong M_2(K).$$

Proof. Passing through Cartier theory and inverting π gives

$$\begin{aligned} \mathrm{End}_D^0(X) &\cong \mathrm{End}_{E_{\mathcal{O}(L),\,\mathrm{gr},\Pi}}(M)[1/\pi] \\ &\cong \mathrm{End}_{\mathcal{K}}(M_0 \otimes_W \mathcal{K}, V\Pi^{-1}) \\ &\cong \mathrm{End}_{\mathcal{K}}(\mathcal{K}^2, \Phi). \end{aligned}$$

For the standard σ^{-1}-linear operator Φ above, a matrix commutes with Φ precisely when all its entries are fixed by σ. Hence

$$\mathrm{End}_{\mathcal{K}}(\mathcal{K}^2, \Phi) = M_2(\mathcal{K})^{\sigma} = M_2(K).$$

□

7.2 Admissible triples

Without further explanation, we will always use X to denote a special formal $\mathcal{O}_D$-module and M its corresponding graded Cartier $\mathcal{O}[\Pi]-$module.

Definition 7.8. An *admissible triple* is a triple (η, T, u) consisting of

1. a graded $\mathcal{O}[\Pi]$-module $\eta = \eta_0 \oplus \eta_1$;
2. a graded $L[\Pi]$-module $T = T_0 \oplus T_1$ with $\dim_L T_0 = \dim_L T_1 = 1$;
3. a homogeneous $\mathcal{O}[\Pi]$-linear map $u\colon \eta \to T$ of degree 0,

satisfying the following conditions:

1. each η_i is a free $\mathcal{O}$-module of rank 2, and the induced map $\eta_i \otimes_{\mathcal{O}} L \longrightarrow T_i$ is surjective;
2. the map induced by u, $\eta/\Pi\eta \longrightarrow T/\Pi T$, is injective.

Proposition 7.9. *For a height-4 special formal $\mathcal{O}_D$-module X, the triple (η_M, T_M, u_M) is admissible.*

Proof. We will first prove they satisfy the condition 1.

If i is critical, we have proved that $\eta_{M,i} \cong M_i^{V^{-1}\Pi}$, and u_i coincides with the composite

$$M_i^{V^{-1}\Pi} \hookrightarrow M_i \longrightarrow M_i/VM_{i-1}.$$

[7] In general, we know if $(\mathcal{K}^2, \Phi)$ is an isocrystal such that $\Phi|_{\mathcal{L}}$ is an isomorphism for some lattice $\mathcal{L}$, then $(\mathcal{K}^2, \Phi)$ is isomorphic to the standard one.

Since $V^{-1}\Pi\colon M_i \to M_i$ is bijective, the pair $(M_i, V^{-1}\Pi)$ is a crystal whose σ^{-1}-linear operator is bijective. Therefore, $(M_i, V^{-1}\Pi) \cong \big(\mathcal{W}^{\oplus 2}, \Phi\big)$, the standard crystal. Consequently,

$$M_i^{V^{-1}\Pi} \cong \big(\mathcal{W}^{\oplus 2}\big)^{\Phi} = (\mathcal{W}^{\sigma})^{\oplus 2} = \mathcal{O}^{\oplus 2}.$$

Moreover, the map

$$M_i^{V^{-1}\Pi} \otimes_{\mathcal{O}} L \longrightarrow M_i/VM_{i-1}$$

is identified with a surjective map $\mathcal{O}^{\oplus 2} \otimes_{\mathcal{O}} L \longrightarrow L$.

Suppose now that i is not critical. Consider the commutative diagram

$$\begin{array}{ccc} \eta_{M,i} & \xrightarrow[\sim]{\Pi} & \eta_{M,i+1} \\ {\scriptstyle u_i}\big\downarrow & & \big\downarrow{\scriptstyle u_{i+1}} \\ T_{M,i} & \xrightarrow[\sim]{\Pi} & T_{M,i+1}. \end{array}$$

The upper horizontal map is an isomorphism by the preceding result. Since $i+1$ is critical, the map u_{i+1} is surjective. The lower horizontal map is also an isomorphism because i is not critical. It follows from the commutativity of the diagram that u_i is surjective.

It remains to prove injectivity of $\eta_M/\Pi\eta_M \to T_M/\Pi T_M$. We will prove $\eta_{M,i}/\Pi\eta_{M,i-1} \to T_{M,i}/\Pi T_{M,i-1}$ is injective. There are three cases.

Case 1: i is critical and $i-1$ is not critical. Then $\Pi(\eta_{M,i-1}) = \eta_{M,i}$, and the injectivity is trivial.

Case 2: both indices are critical. Consider the commutative diagram

$$\begin{array}{ccc} \eta_{M,i-1} & \xrightarrow{\Pi} & \eta_{M,i} \\ \big\downarrow{\scriptstyle \cong} & & \big\downarrow{\scriptstyle \cong} \\ M_{i-1}^{V^{-1}\Pi} & \xrightarrow{\Pi} & M_i^{V^{-1}\Pi} \\ \big\downarrow & & \big\downarrow \\ T_{M,i-1} & \xrightarrow[\Pi=0]{} & T_{M,i}. \end{array}$$

Let $x \in \eta_{M,i}$ and suppose that its image in

$$T_{M,i}/\Pi T_{M,i-1} = T_{M,i}$$

is zero. Then $x = Vy$ for some $y \in M_{i-1}$. Since $x \in M_i^{V^{-1}\Pi}$, we have $\Pi x = Vx$. Therefore,

$$V\Pi y = \Pi V y = \Pi x = Vx = VVy.$$

Since V is injective, it follows that $\Pi y = Vy$. Thus $y \in M_{i-1}^{V^{-1}\Pi} = \eta_{M,i-1}$, and hence $x = Vy = \Pi y$. Therefore, the class of x in $\eta_{M,i}/\Pi\eta_{M,i-1}$ is zero.

Case 3: i is not critical and $i-1$ is critical. We have a commutative diagram

$$\begin{array}{ccccc} \eta_{M,i-1} & \xrightarrow{\Pi} & \eta_{M,i} & \xrightarrow[\sim]{\Pi} & \eta_{M,i-1} \\ {\scriptstyle u_{i-1}}\big\downarrow & & {\scriptstyle u_i}\big\downarrow & & \big\downarrow{\scriptstyle u_{i-1}} \\ T_{M,i-1} & \xrightarrow[\Pi=0]{} & T_{M,i} & \xrightarrow[\sim]{\Pi} & T_{M,i-1}. \end{array}$$

Let $x \in \eta_{M,i}$ and suppose that

$$u_i(x) = 0 \qquad \text{in} \qquad T_{M,i}/\Pi T_{M,i-1} = T_{M,i}.$$

By the commutativity of the diagram, $u_{i-1}(\Pi x) = 0$. Hence $\Pi x = Vy$ for some $y \in M_i$. Since $\Pi x \in \eta_{M,i-1} = M_{i-1}^{V^{-1}\Pi}$, we have $\Pi(\Pi x) = V(\Pi x)$. Using $\Pi x = Vy$ and the injectivity of V, we obtain $\Pi y = Vy$. Thus $y \in M_i^{V^{-1}\Pi}$.

Moreover, $\overline{\Pi y} = \overline{Vy}$ is vanish in $T_{M,i-1}$. Since $\Pi\colon T_{M,i} \xrightarrow{\sim} T_{M,i-1}$ is an isomorphism, it follows that $\overline{y} = 0$ in $T_{M,i}$. Therefore, $y = Vz$ for some $z \in M_{i-1}$. The relation $\Pi y = Vy$ and the injectivity of V imply $\Pi z = Vz$, so $z \in M_{i-1}^{V^{-1}\Pi} = \eta_{M,i-1}$. Finally,

$$\Pi x = Vy = V^2 z = \Pi^2 z.$$

Since Π is injective, we conclude that $x = \Pi z$. Thus the class of x in $\eta_{M,i}/\Pi\eta_{M,i-1}$ is zero. □

Let $\mathscr{S}$ be the groupoid of height-4 special formal $\mathcal{O}_D$-modules over L, and let $\mathscr{A}$ be the groupoid of admissible triples. In this discussion, we ignore automorphisms and regard these groupoids as sets of isomorphism classes. There is a functor

$$\mathcal{F} : \mathscr{S} \longrightarrow \mathscr{A}, \qquad X \longmapsto (\eta_M, T_M, u_M).$$

Proposition 7.10. *The functor $\mathcal{F}$ is fully faithful.*

Proof. We show that M_0, M_1, Π, and V are determined by (η, T, u).

Suppose that i is critical. The inclusion $\eta_i \cong M_i^{V^{-1}\Pi} \hookrightarrow M_i$ induces an isomorphism of crystals

$$\varphi\colon \big(\eta_i \otimes_{\mathcal{O}} \mathcal{W}, \mathrm{id} \otimes \sigma^{-1}\big) \xrightarrow{\sim} (M_i, \Pi V^{-1}).$$

Let $H = \ker\big(u_i \otimes \mathrm{id}\colon \eta_i \otimes_{\mathcal{O}} \mathcal{W} \to T_i\big)$. Then $\varphi(H) = VM_{i-1}$ and $\varphi(\sigma(H)) = \Pi M_{i-1}$.

Thus, the diagram (whose vertical arrows are isomorphisms)

$$\begin{array}{ccccc}
M_{i-1} & \underset{V}{\overset{\Pi}{\rightleftarrows}} & M_i & \underset{V}{\overset{\Pi}{\rightleftarrows}} & M_{i+1} \\
{\scriptstyle \Pi^{-1}\circ\varphi}\uparrow & & \uparrow{\scriptstyle \varphi} & & \uparrow{\scriptstyle \Pi^{-1}\circ\varphi} \\
\sigma(H) & \underset{incl\circ\sigma^{-1}}{\overset{incl}{\rightrightarrows}} & \eta_i \otimes \mathcal{W} & \underset{\pi\circ\sigma^{-1}}{\overset{\pi}{\rightrightarrows}} & \sigma(H)
\end{array}$$

identifies the Cartier datum with data determined by (η, T, u).

It follows that every isomorphism $\mathcal{F}(X) \xrightarrow{\sim} \mathcal{F}(X')$ uniquely determines an isomorphism $M \xrightarrow{\sim} M'$, and therefore an isomorphism $X \xrightarrow{\sim} X'$.

Therefore, $\mathcal{F}$ is fully faithful. □

Remark 7.11. It is unnecessary to reconstruct F separately, since $F = V^{-1} \circ \pi$.

Proposition 7.12. *The functor $\mathcal{F}$ is essentially surjective.*

Proof. Let (η, T, u) be an admissible triple. We construct a height-4 special formal $\mathcal{O}_D$-module X such that $(\eta_M, T_M, u_M) \cong (\eta, T, u)$.

Choose a critical index i and set $H = \ker\big(u_i \otimes \mathrm{id}\colon \eta_i \otimes_{\mathcal{O}} \mathcal{W} \to T_i\big)$. Define $M_i = \eta_i \otimes_{\mathcal{O}} \mathcal{W}$ and $M_{i-1} = \sigma(H)$. Define Π and V by the following diagram.

$$\begin{array}{ccccc}
M_{i-1} & \underset{V}{\overset{\Pi}{\rightleftarrows}} & M_i & \underset{V}{\overset{\Pi}{\rightleftarrows}} & M_{i+1} \\
{\scriptstyle \Pi^{-1}\circ\varphi}\uparrow & & \uparrow{\scriptstyle \varphi} & & \uparrow{\scriptstyle \Pi^{-1}\circ\varphi} \\
\sigma(H) & \underset{incl\circ\sigma^{-1}}{\overset{incl}{\rightrightarrows}} & \eta_i \otimes \mathcal{W} & \underset{\pi\circ\sigma^{-1}}{\overset{\pi}{\rightrightarrows}} & \sigma(H)
\end{array}$$

It remains to verify that $[M_i : VM_{i-1}] = [M_{i-1} : VM_i] = [M_i : \Pi M_{i-1}] = [M_{i-1} : \Pi M_i] = 1$.

By condition 2 in the definition of an admissible triple, the map $\eta_i \otimes_{\mathcal{O}} L \to T_i$ is surjective. Hence $(\eta_i \otimes_{\mathcal{O}} \mathcal{W})/H \cong T_i \cong L$. Therefore, $\pi(\eta_i \otimes_{\mathcal{O}} \mathcal{W}) \subsetneq H \subsetneq \eta_i \otimes_{\mathcal{O}} \mathcal{W}$. The same statement holds for $\sigma(H)$.

Since $\eta_i \otimes_{\mathcal{O}} \mathcal{W} \cong \mathcal{W}^{\oplus 2}$, we have $[\eta_i \otimes_{\mathcal{O}} \mathcal{W} : \pi(\eta_i \otimes_{\mathcal{O}} \mathcal{W})] = 2$. Thus, all four indices above are equal to 1.

Moreover,

$$\big(M_i^{V^{-1}\Pi}, M_i/VM_{i-1}, \mathrm{can}\big) = \big(\eta_i, (\eta_i \otimes_{\mathcal{O}} \mathcal{W})/H, \mathrm{can}\big) \cong (\eta_i, T_i, u_i).$$

If X is the special formal $\mathcal{O}_D$-module corresponding to M, then $\mathcal{F}(X) \cong (\eta, T, u)$ by the following lemma. □

Lemma 7.13. *Let (η, T, u) be an admissible triple, and suppose that i is critical. Then (η, T, u) is determined, up to isomorphism, by (η_i, T_i, u_i).*

Proof. Put $H = \ker(u_i \colon \eta_i \to T_i)$. Since u_i is surjective after tensoring with L, we have $\pi\eta_i \subseteq H \subsetneq \eta_i$.

First suppose that $i-1$ is critical. Then $\Pi \colon T_{i-1} \to T_i$ is zero. By condition 2, the map $\eta_i/\Pi\eta_{i-1} \to T_i/\Pi T_{i-1} = T_i$ is injective. Consequently, $\Pi \colon \eta_{i-1} \to \eta_i$ factors through H, so $\pi\eta_i \subseteq \Pi\eta_{i-1} \subseteq H \subsetneq \eta_i$.

Similarly, $\Pi\eta_i \subsetneq \eta_{i-1}$. The induced map

$$\eta/\Pi\eta \otimes_{\mathcal{O}} L \longrightarrow T/\Pi T$$

is an isomorphism by condition 2 and checking dimensions. Hence $\eta_i/\Pi\eta_{i-1} \to T_i$ is an isomorphism after tensoring with L, and therefore $H \subseteq \Pi\eta_{i-1}$.

Thus, $H = \Pi\eta_{i-1}$. Moreover, $T_{i-1} \cong \eta_{i-1}/\Pi\eta_i \otimes_{\mathcal{O}} L$, and u_{i-1} is the composite

$$\eta_{i-1} \longrightarrow \eta_{i-1}/\Pi\eta_i \longrightarrow \eta_{i-1}/\Pi\eta_i \otimes_{\mathcal{O}} L \xrightarrow{\sim} T_{i-1}.$$

Therefore, (η, T, u) is determined.

Now suppose that $i-1$ is not critical. Then $\Pi \colon T_{i-1} \xrightarrow{\sim} T_i$. It follows that $\Pi \colon \eta_{i-1} \xrightarrow{\sim} \eta_i$, so the whole triple is again determined by (η_i, T_i, u_i).

Indeed, let $x \in \eta_i$ with $u_i(x) = 0$. By the commutative diagram

$$\begin{array}{ccccc}
\eta_i & \xrightarrow{\Pi} & \eta_{i-1} & \xrightarrow[\cong]{\Pi} & \eta_i \\
{\scriptstyle u_i}\downarrow & & {\scriptstyle u_{i-1}}\downarrow & & \downarrow{\scriptstyle u_i} \\
T_i & \xrightarrow[\Pi=0]{} & T_{i-1} & \xrightarrow[\cong]{\Pi} & T_i,
\end{array}$$

there exists $y \in \eta_{i-1}$ such that $x = \Pi y$ and $u_{i-1}(y) = 0$.

Since $\eta_{i-1}/\Pi\eta_i \to T_{i-1}/\Pi T_i = T_{i-1}$ is injective, we have $y \in \Pi\eta_i$. Hence $x \in \Pi^2\eta_i = \pi\eta_i$. Thus, $H \subseteq \pi\eta_i$, and therefore $H = \pi\eta_i$.

We have consequently shown that $i-1$ is not critical precisely when $H = \pi\eta_i$. Hence the criticality of $i-1$, and thus the entire triple, is determined by (η_i, T_i, u_i). □

Therefore, we have the following theorem.

Theorem 7.14. *The functor $\mathcal{F}$ is an equivalence of categories.*

7.3 The level structure

Although we have shown that the groupoid $\mathscr{S}$ is equivalent to a groupoid consisting of "linear algebra" objects, $\mathscr{A}$, it is still unknown whether $\mathscr{A}$ is representable by some geometric space. But $\mathscr{A}$ is closely related to $F(L)$, where F is the moduli functor representable by the p-adic upper plane $\widehat{\Omega}$, and the remaining issue will be resolved by imposing a suitable level structure.

Lemma 7.15. *Suppose that $\mathcal{F}(X) = (\eta, T, u)$. For $i = 0, 1$, the natural map*

$$\eta_i \otimes_{\mathcal{O}} \mathcal{K} \longrightarrow M_i \otimes_{\mathcal{W}} \mathcal{K}, \qquad x \otimes t \longmapsto x \otimes t,$$

is an isomorphism of isocrystals

$$\big(\eta_i \otimes_{\mathcal{O}} \mathcal{K}, \mathrm{id} \otimes \sigma^{-1}\big) \xrightarrow{\sim} \big(M_i \otimes_{\mathcal{W}} \mathcal{K}, V^{-1}\Pi\big).$$

Proof. Since $\eta_i \cong M_i^{V^{-1}\Pi}$, we have $\eta_i \otimes_{\mathcal{O}} K \cong (M_i \otimes_{\mathcal{W}} \mathcal{K})^{V^{-1}\Pi}$.

We also know that $(M_i \otimes_{\mathcal{W}} \mathcal{K}, V^{-1}\Pi) \cong (\mathcal{K}^{\oplus 2}, \Phi)$, where $\Phi(ae_1 + be_2) = \sigma^{-1}(a)e_1 + \sigma^{-1}(b)e_2$. It is therefore enough to observe that $(\mathcal{O}^{\oplus 2} \otimes_{\mathcal{O}} \mathcal{K}, \mathrm{id} \otimes \sigma^{-1}) \cong (\mathcal{K}^{\oplus 2}, \Phi)$. □

Definition 7.16. Define $\mathrm{QIsog}(X, X') \subseteq \mathrm{Hom}_{\mathcal{O}_D}(X, X') \otimes_{\mathcal{O}} K$ as follows. An element α belongs to $\mathrm{QIsog}(X, X')$ if there exists $n \in \mathbb{N}$ such that $\pi^n \alpha \colon X \to X'$ is an $\mathcal{O}_D$-isogeny.

Define the height of α by $\mathrm{ht}(\alpha) = \mathrm{ht}(\pi^n \alpha) - \mathrm{ht}(\pi^n)$.

Proposition 7.17. *There is a natural identification*

$$\mathrm{QIsog}(X, X') \cong \mathrm{Isom}_{\mathrm{isoc}}\big((M, V^{-1}\Pi), (M', V^{-1}\Pi)\big).$$

Proof. For any $\alpha \in \mathrm{QIsog}(X, X')$, there exists $n \in \mathbb{N}$ such that $\pi^n \alpha$ is an isogeny. Thus, $\pi^n \alpha$ induces an isomorphism $\beta \colon (M, V^{-1}\Pi) \to (M', V^{-1}\Pi)$. Define a map by sending α to $\pi^{-n}\beta$. One checks that this definition is independent of the choice of n. We can define an inverse by a similar argument in Theorem 7.5. □

Proposition 7.18. *There is a natural bijection*

$$\mathrm{QIsog}(X, X') \cong \mathrm{Isom}_K\big(\eta_0 \otimes_{\mathcal{O}} K, \eta_0' \otimes_{\mathcal{O}} K\big).$$

Proof. By the preceding results,

$$\begin{aligned}
\mathrm{QIsog}(X, X') &\cong \mathrm{Isom}_{\mathrm{isoc}}\big((M, V^{-1}\Pi), (M', V^{-1}\Pi)\big) \\
&\cong \mathrm{Isom}_{\mathrm{isoc}}\big((M_0, V^{-1}\Pi), (M_0', V^{-1}\Pi)\big) \\
&\cong \mathrm{Isom}_{\mathrm{isoc}}\big((\eta_0 \otimes_{\mathcal{O}} \mathcal{K}, \mathrm{id} \otimes \sigma^{-1}), (\eta_0' \otimes_{\mathcal{O}} \mathcal{K}, \mathrm{id} \otimes \sigma^{-1})\big) \\
&\cong \mathrm{Isom}_K\big(\eta_0 \otimes_{\mathcal{O}} K, \eta_0' \otimes_{\mathcal{O}} K\big).
\end{aligned}$$

□

Proposition 7.19. *Suppose that 0 is critical for X. Then every height-zero quasi-isogeny $\alpha \colon X \to X'$ corresponds to an isomorphism $\rho \colon \eta_0 \otimes_{\mathcal{O}} K \xrightarrow{\sim} \eta_0' \otimes_{\mathcal{O}} K$ satisfying*

$$\begin{cases} [\eta_0' : \rho(\eta_0)] = 0, & \text{if } 0 \text{ is critical for } X', \\ [\eta_1' : \Pi\rho(\eta_0)] = 1, & \text{if } 1 \text{ is critical for } X'. \end{cases} \tag{$*$}$$

Proof. Recall that if $M, M' \supseteq N$, then $[M : M'] := [M : N] - [M' : N]$.

By the preceding proposition, there exists an isomorphism $\rho\colon \eta_0 \otimes_{\mathcal{O}} K \xrightarrow{\sim} \eta_0' \otimes_{\mathcal{O}} K$ corresponding to $\alpha \in \mathrm{QIsog}(X, X')$. Since $\mathrm{ht}(\alpha) = 0$, we have

$$\begin{aligned}[M' : \alpha(M)] &= [M' : \alpha(\pi^n M)] - [\alpha(M) : \alpha(\pi^n M')] \\ &= \mathrm{ht}(\pi^n \alpha) - [M : \pi^n M] \\ &= \mathrm{ht}(\pi^n) - \mathrm{ht}(\pi^n) = 0.\end{aligned}$$

Moreover, $[M' : \alpha(M)] = 0$ if and only if $[M_0' : \alpha(M_0)] + [M_1' : \alpha(M_1)] = 0$.

If 0 is critical for X', then

$$\begin{aligned}0 &= [M_0' : \alpha(M_0)] + [M_1' : \alpha(M_1)] \\ &= [M_0' : \alpha(M_0)] + [M_1' : \Pi M_0'] + [\Pi M_0' : \Pi\alpha(M_0)] + [\Pi\alpha(M_0) : \alpha(M_1)] \\ &= 2[M_0' : \alpha(M_0)].\end{aligned}$$

Hence $[M_0' : \alpha(M_0)] = 0$. Since $M_0 \cong \eta_0 \otimes_{\mathcal{O}} \mathcal{W}$ and $M_0' \cong \eta_0' \otimes_{\mathcal{O}} \mathcal{W}$, this is equivalent to $[\eta_0' : \rho(\eta_0)] = 0$.

If 1 is critical for X', then

$$\begin{aligned}0 &= [M_0' : \alpha(M_0)] + [M_1' : \alpha(M_1)] \\ &= [M_0' : \Pi M_1'] + [\Pi M_1' : \Pi\alpha(M_1)] + [\Pi\alpha(M_1) : \alpha(M_0)] + [M_1' : \alpha(M_1)].\end{aligned}$$

It follows that $[M_1' : \alpha(M_1)] = 0$. Therefore,

$$[M_1' : \Pi\alpha(M_0)] = [\alpha(M_1) : \Pi\alpha(M_0)] = [M_1 : \Pi M_0] = 1.$$

Since $M_1' \cong \eta_1' \otimes_{\mathcal{O}} \mathcal{W}$, this is equivalent to $[\eta_1' : \Pi\rho(\eta_0)] = 1$. □

Remark 7.20. Condition (*) implies that $\bigwedge^2 \eta_i' = \pi^{-i} \bigwedge^2 \Pi^i \rho(\eta_0)$ whenever $i \in \{0, 1\}$ is critical for X'.

Recall that Φ is the height-4 special formal $\mathcal{O}_D$-module over k that was fixed previously. Fix an isomorhism $\mathcal{O}^2 \cong \eta_{\Phi,0}$. Consider the groupoid of pairs (X, ρ), where X is a height-4 special formal $\mathcal{O}_D$-module over L and $\rho\colon \Phi_L \to X$ is a height-zero quasi-isogeny.

By the preceding results, this groupoid is equivalent to the groupoid of quadruples (η, T, u, r), where (η, T, u) is an admissible triple and $r\colon K^2 \xrightarrow{\sim} \eta_0 \otimes_{\mathcal{O}} K$ satisfies

$$\bigwedge^2 \eta_i = \pi^{-i} \bigwedge^2 \Pi^i r(\mathcal{O}^2)$$

whenever i is critical.

This is precisely $\widehat{\Omega}(L)$.

8 Filtrations and the sheaf-theoretic viewpoint

In this section, we assume that B is an $\mathcal{O}'$-algebra such that $\pi B = 0$, and let M be a graded special Cartier $\mathcal{O}[\Pi]$-module. Fix a critical index $i \in \{0, 1\}$; thus $\Pi M_i \subseteq V M_i$. Write $N = N(M)$, $\phi = \phi_M$, and $\eta = \eta_M$.

8.1 Filtrations

The $\mathcal{O}$-linear inclusion ι_2 induces maps $\iota_2\colon V^{2n}M_i \hookrightarrow N_i$, $V^{2n}m \mapsto \overline{(0, V^{2n}m)}$, and $\iota_2\colon V^{2n-1}M_i \hookrightarrow N_{i-1}$, $V^{2n-1}m \mapsto \overline{(0, V^{2n-1}m)}$.

Set $N_{i,n} = N_i/V^{2n}M_i$ and $N_{i-1,n} = N_{i-1}/V^{2n-1}M_i$. More generally, for $j \in \{0,1\}$, define $\epsilon = \epsilon(j) = \delta_{j,i-1}$. Then $N_{j,n} = N_j/V^{2n-\epsilon}M_i$.

Lemma 8.1. *The endomorphism $\phi \circ \iota_2$ acts $\mathcal{O}$-linearly on V^rM for every $r \geq 1$.*

Proof. For $V^rm \in V^rM$, we have

$$(\phi \circ \iota_2)(V^rm) = \phi(\overline{(0, V^rm)}) = \overline{(V^rm, 0)} = \overline{(0, \Pi V^{r-1}m)}.$$

□

Define $\eta_{j,n} := N_{j,n}^{\phi}$.

Lemma 8.2. *There are natural identifications $N_j \simeq \varprojlim_n N_{j,n}$ and $\eta_j \simeq \varprojlim_n \eta_{j,n}$.*

Proof. By definition, there is an exact sequence

$$0 \longrightarrow M_{j-1} \xrightarrow{\alpha=(V,-\Pi)} M_j \oplus M_j \longrightarrow N_j \longrightarrow 0.$$

Moreover, $\alpha(M_{j-1}) \cap (0, V^{2n-\epsilon}M_i) = 0$. Consequently, the snake lemma applied to

$$\begin{array}{ccccccccc} 0 & \longrightarrow & 0 & \longrightarrow & V^{2n-\epsilon}M_i & \xrightarrow{\iota_2} & (0, V^{2n-\epsilon}M_i) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & M_{j-1} & \xrightarrow{\alpha} & M_j \oplus M_j & \longrightarrow & N_j & \longrightarrow & 0 \end{array}$$

gives an exact sequence

$$0 \longrightarrow M_{j-1} \longrightarrow \frac{M_j \oplus M_j}{(0, V^{2n-\epsilon}M_i)} \longrightarrow N_{j,n} \longrightarrow 0.$$

Apply $\varprojlim$ to the above sequence, we obtain the following exact sequence by Mittag-Leffler lemma:

$$0 \longrightarrow M_{j-1} \xrightarrow{\alpha} M_j \oplus M_j \longrightarrow \varprojlim_n N_{j,n} \longrightarrow 0.$$

Hence $\varprojlim_n N_{j,n} \simeq N_j$.

The functor $\varprojlim$ is left exact. Applying it to $0 \to \eta_{j,n} \to N_{j,n} \xrightarrow{1-\phi} N_{j,n}$ gives $\varprojlim_n \eta_{j,n} = N_j^{\phi} = \eta_j$. □

8.2 Sheaf-version

We now regard M_j, N_j, η_j, $N_{j,n}$, and $\eta_{j,n}$ as functors on Alg_B with values in $\mathrm{Mod}_{\mathcal{O}}$. For every B-algebra B', set

$$M(B') = E_{\mathcal{O}}(B')\widehat{\otimes}_{E_{\mathcal{O}}(B)}M.$$

The functors $N_j(B')$, $\eta_j(B')$, $N_{j,n}(B')$, and $\eta_{j,n}(B')$ are defined in the same manner. Their transition morphisms are well defined because ϕ is compatible with base change.

Lemma 8.3. *As a functor on* Alg_B*, one has* $N_{j,n} \simeq \mathbb{A}_B^{2n+1-\epsilon}$.

Proof. Choose a homogeneous V-basis $\{\gamma_0, \gamma_1\}$ of M. Define

$$M_{j,(0)}(B') := \{[a]\gamma_j \in M(B') \mid a \in B'\}.$$

This construction is compatible with base change. There is a natural map

$$M_{j,(0)} \times M_j/V^{2n-\epsilon}M_i \longrightarrow N_{j,n}.$$

We first prove this map is surjective. Let $\overline{(m, m')} \in N_j$. Write $m = m_0 + Vm_1$, where $m_0 \in M_{j,(0)}$ and $m_1 \in M_{j-1}$. Then

$$\overline{(m, m')} = \overline{(m_0, m' + \Pi m_1)},$$

so every element of $N_{j,n}$ lies in the image.

Then we will prove that the map is injective. Suppose $\overline{(m_0, m')} \equiv \overline{(\ell_0, \ell')}$ in $N_{j,n}$ for some $m_0, l_0 \in M_{j,(0)}$ and $m', l' \in M_j$. Then there exists $t \in M_{j-1}$ such that

$$m_0 + Vt = \ell_0 \text{ and } m' - \Pi t \equiv \ell' \pmod{V^{2n-\epsilon}M_i}.$$

The uniqueness of the V-adic expansion implies $t = 0$. Hence

$$(m_0, m') \equiv (\ell_0, \ell') \mod 0 \times V^{2n-\epsilon}M_i$$

already in the source.

Thus, for every B-algebra B', there is a natural bijection

$$\mathbb{A}_B^{2n+1-\epsilon}(B') \xrightarrow{\sim} N_{j,n}(B'),$$

given explicitly by

$$(a, a_0, \ldots, a_{2n-\epsilon-1}) \longmapsto \left([a]\gamma_j, \sum_{0 \le k < 2n-\epsilon} V^k[a_k]\gamma_{j-k}\right).$$

□

Lemma 8.4. *The functor* $\eta_{j,n}$ *is represented by a finitely presented* B*-algebra.*

Proof. The map $1 - \phi \colon N_{j,n} \to N_{j,n}$ is a morphism of functors. Write $N = 2n + 1 - \epsilon$. By the preceding lemma, $N_{j,n} \simeq \operatorname{Spec} B[X_1, \ldots, X_N]$.

Let $P_i \in B[X_1, \ldots, X_N]$ be the image of X_i under the B-algebra homomorphism corresponding to $1 - \phi$. Since $\eta_{j,n}$ is the fiber of $1 - \phi$ over the zero section, it is represented by

$$\operatorname{Spec}\left(B[X_1, \ldots, X_N]/(P_1, \ldots, P_N)\right).$$

In particular, its representing algebra is finitely presented over B. □

Lemma 8.5. *Let* R *be a finitely presented* B*-algebra. Suppose that, for every surjection of* B*-algebras* $B' \twoheadrightarrow B''$ *with square-zero kernel, the map* $\operatorname{Spec}(R)(B') \to \operatorname{Spec}(R)(B'')$ *is bijective. Then* $\operatorname{Spec} R$ *is étale over* B.

Proof. We first prove that $\Omega_{R/B} = 0$. Let $I = \ker(R \otimes_B R \to R)$. Then $\Omega_{R/B} \simeq I/I^2$. The morphism $(R \otimes_B R)/I^2 \twoheadrightarrow R$ has square-zero kernel. Therefore,

$$\mathrm{Hom}_B\big(R, (R \otimes_B R)/I^2\big) \xrightarrow{\sim} \mathrm{Hom}_B(R, R).$$

Consider the two morphisms $\iota_1, \iota_2 \colon R \to (R \otimes_B R)/I^2$ defined by $\iota_1(r) = r \otimes 1$ and $\iota_2(r) = 1 \otimes r$. Their composites with the multiplication map to R coincide, so $\iota_1 = \iota_2$. Hence $r \otimes 1 - 1 \otimes r \in I^2$ for every $r \in R$. Thus $dr = 0$ in I/I^2, and consequently $\Omega_{R/B} = 0$.

The infinitesimal lifting property also implies that R is flat over B. The proof of this fact is not straightforward, and we use it without proof. Since R is finitely presented, flat, and unramified over B, it is étale over B. □

Corollary 8.6. *The B-scheme $\eta_{j,n}$ is étale over B.*

Proof. It suffices to verify the infinitesimal lifting property. Let $B' \twoheadrightarrow B''$ be a surjective morphism of B-algebras with nilpotent kernel. We have surjective morphisms:

$$\alpha \colon M_i(B') \to M_i(B'') \text{ and } \beta \colon N_j(B') \to N_j(B'').$$

They induce a morphism $\gamma \colon N_{j,n}(B') \to N_{j,n}(B'')$. We have the following commutative diagram:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & V^{2n-\epsilon}M_i(B') & \longrightarrow & N_j(B') & \longrightarrow & N_{j,n}(B') & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\
0 & \longrightarrow & V^{2n-\epsilon}M_i(B'') & \longrightarrow & N_j(B'') & \longrightarrow & N_{j,n}(B'') & \longrightarrow & 0.
\end{array}$$

The snake lemma shows that γ is surjective. Since ϕ acts nilpotently on $\ker\beta$ by Proposition 5.8, the same holds for $\ker\gamma$. Thus, the endomorphism $1 - \phi$ is invertible on $\ker\gamma$. Applying the snake lemma to

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \ker\gamma & \longrightarrow & N_{j,n}(B') & \xrightarrow{\gamma} & N_{j,n}(B'') & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle 1-\phi} & & \downarrow{\scriptstyle 1-\phi} & & \downarrow{\scriptstyle 1-\phi} & & \\
0 & \longrightarrow & \ker\gamma & \longrightarrow & N_{j,n}(B') & \xrightarrow{\gamma} & N_{j,n}(B'') & \longrightarrow & 0
\end{array}$$

gives a bijection $\eta_{j,n}(B') \xrightarrow{\sim} \eta_{j,n}(B'')$. The preceding lemma therefore shows that $\eta_{j,n}$ is étale over B. □

Let $S = \mathrm{Spec}\, B$. Let S_{et} be the small étale site on S.

Corollary 8.7. *The functor $\eta_{j,n}$ defines an étale sheaf on S_{et}.*

Proof. This follows from the fact that $\eta_{j,n}$ is represented by an étale S-scheme. □

Corollary 8.8. *For any morphism of schemes $f : S' \to S$, we have $f^*\eta_{j,n}(S') = \eta_{j,n}(S')$.*

Proof. One has $f^*\eta_{j,n} = \mathrm{Hom}_{S'_{\mathrm{et}}}(-, \eta_{j,n} \times_S S')$. Consequently,

$$(f^*\overline{\eta}_{j,n})(S') = \mathrm{Hom}_{S'}(S', \eta_{j,n} \times_S S') = \mathrm{Hom}_S(S', \eta_{j,n}) = \eta_{j,n}(S').$$

□

Corollary 8.9. *The functor η_j is a sheaf on S_{et}.*

We extend η_j to a functor on S_{et} by using $\eta_j(V) \simeq \varprojlim_n \eta_{j,n}(V)$ for any $V \in S_{\mathrm{et}}$. It is compatible with the original definition of η_j.

Proof. Let $\{U_k \to U\}_k$ be an étale covering in S_{et}. For every n, the sheaf property of $\eta_{j,n}$ gives an exact sequence

$$0 \longrightarrow \eta_{j,n}(U) \longrightarrow \prod_k \eta_{j,n}(U_k) \longrightarrow \prod_{k,\ell} \eta_{j,n}(U_k \times_U U_\ell).$$

Apply $\varprojlim$. Since the inverse-limit functor is left exact and commutes with arbitrary products. , we obtain the exact sequence

$$0 \longrightarrow \eta_j(U) \longrightarrow \prod_k \eta_j(U_k) \longrightarrow \prod_{k,\ell} \eta_j(U_k \times_U U_\ell).$$

Therefore η_j is a sheaf on S_{et}. □

8.3 More properties

We will first recall some knowledge of étale cohomology.

Definition 8.10. A sheaf $\mathscr{F}$ of $\mathcal{O}$-modules on S is called *constructible* if, for every affine open subscheme $U \subseteq S$, there exists a finite constructible decomposition $U = \coprod_i U_i$ such that each restriction $\mathscr{F}|_{U_i}$ is locally constant and finitely generated.

Here each $|U_i|$ is constructible; equivalently, it is a finite union of subsets of the form $U' \cap V'^{\mathrm{c}}$, where U' and V' are open and quasi-compact.[8]

Corollary 8.11. *The sheaf $\eta_{j,n}$ is constructible.*

Proof. This follows from the standard constructibility theorem for étale morphisms of finite presentation. □

Definition 8.12. Let X be a general scheme. Let $\mathscr{F}$ be a sheaf of A-modules on X_{et}. We say that $\mathscr{F}$ is smooth if $\mathscr{F}$ is locally constant for the étale topology. To be explicit, for any geometric point x, there exists an étale neighborhood of x, U, such that $\mathscr{F}|_{U_{\mathrm{et}}}$ is isomorphic to the constant sheaf $\underline{M}$ for some finite presented A-modules M.

Lemma 8.13. *If the sheaf $\mathscr{F}$ is represented by a scheme Y which is finite étale over X, then $\mathscr{F}$ is smooth.*

Proof. For any geometric point $\bar{x} \to X$, there exists an étale neighborhood $U \to X$ of $\bar{x}$ such that

$$Y_U \cong \bigsqcup_{i \in I} U$$

for some finite set I, since Y is finite étale over X. Therefore,

$$\mathscr{F}|_{U_{\mathrm{et}}} \cong \underline{I},$$

and hence $\mathscr{F}$ is smooth. □

Lemma 8.14. *Let $f\colon Y \to X$ be a separated étale morphism. Assume that $\#Y_{\bar{x}} = n \in \mathbb{N}_+$ for every geometric point $\bar{x} \to X$. Then f is finite étale.*

[8] In general, the definition is more complicated. In our setting, the Noetherianity of $\mathcal{O}$ and the affineness of S allow us to simplify the definition.

Sketch of the proof. Fix a geometric point $\bar{x} \to X$ and write $Y_{\bar{x}} = \{\bar{y}_1, \ldots, \bar{y}_n\}$. Since f is étale, after replacing X by an étale neighborhood $U \to X$ of $\bar{x}$, we may find sections

$$s_1, \ldots, s_n \colon U \to Y_U$$

lifting $\bar{y}_1, \ldots, \bar{y}_n$, respectively.

Since f is unramified, the diagonal

$$\Delta_f \colon Y_U \to Y_U \times_U Y_U$$

is an open immersion, while the separatedness of f implies that it is also a closed immersion. Hence, for $i \neq j$, the coincidence locus

$$E_{i,j} = \{u \in U \mid s_i(u) = s_j(u)\}$$

is open and closed in U. Since $\bar{y}_i \neq \bar{y}_j$, the point $\bar{x}$ does not lie in $E_{i,j}$. After shrinking U around $\bar{x}$, we may therefore assume that $E_{i,j} = \varnothing$ for all $i \neq j$. Thus the sections $s_1, \ldots, s_n$ are pairwise disjoint and give an open and closed immersion

$$\coprod_{i=1}^{n} U \longrightarrow Y_U.$$

This immersion is in fact an isomorphism. Indeed, for every geometric point $\bar{u} \to U$, the n sections give n distinct points of $Y_{\bar{u}}$, while by assumption$\#Y_{\bar{u}} = n$. Hence there are no further geometric points in the complement, so the complement is empty. Therefore Y_U is isomorphic to the disjoint union of n copies of U. Thus f is finite after an étale localization on the target. Since finiteness is local for the étale topology on the target, f is finite. As f is already étale, it is finite étale. □

We now return to our original topic.

It is clear that we could also view T and T_0, T_1 locally free sheaves on S_{et}. Let S_{i-1} be the closed subset of S where $i-1$ is critical, i.e. S_{i_1} is the zero locus of $\Pi : T_{i-1} \to T_i$.

Proposition 8.15. *1. $\eta_{i,n}$ is a smooth sheaf over S of free $\mathcal{O}/\pi^n$-modules of rank 2.*

2. $\eta_{i-1,n}$ is constructible over S and $\Pi \colon \eta_{i-1,n} \longrightarrow \eta_{i,n}$ is injective. Moreover:

- *(a) $\eta_{i-1,n}$ is smooth above $S - S_{i-1}$ and Π is an isomorphism above $S - S_{i-1}$.*
- *(b) $\eta_{i-1,n}$ is smooth over S_{i-1} and $\eta_{i,n}/\Pi\eta_{i-1,n}$ is a smooth sheaf over S_{i-1} of free $\mathcal{O}/\pi$-module of rank 1.*

Proof. The injectivity of a morphism of sheaves is local on the base. By the above lemmas, to prove that the sheaves in question are smooth, it suffices to show that the cardinality of their fibers at geometric points of S is constant. Since $\mathcal{O}/\pi^n$ is finite, the relevant assertions may be checked on geometric fibers. We may therefore assume that $B = L$ is an algebraically closed field of characteristic p.

The proof of 1:By Lemma 6.3 and Lemma 6.4, we have an isomorphism

$$N_{i,n} \cong M_i/VM_{i-1} \oplus M_i/V^{2n}M_i$$

such that the endomorphism ϕ of $N_{i,n}$ corresponds with

$$(\overline{m}, \overline{m''}) \longmapsto (\overline{m''}, V^{-1}\Pi\overline{m''}).$$

We thus have an isomorphism:

$$\eta_{i,n} \cong \left(M_i/V^{2n}M_i\right)^{V^{-1}\Pi}.$$

Identify $(M_i, V^{-1}\Pi)$ with $(\eta_i \otimes_{\mathcal{O}} \mathcal{W}, \sigma)$; then V^2 is identified with $\pi \cdot \sigma^{-2}$ and $V^{2n}M_i$ with $\pi^n\eta_i \otimes_{\mathcal{O}} \mathcal{W}$; thus

$$\eta_{i,n} \cong \eta_i/\pi^n\eta_i \cong (\mathcal{O}/\pi^n)^2.$$

The proof of 2*:* The map $\Pi\colon M_{i-1} \to M_i$ is injective[9] and $\Pi N_{i-1} \cap V^{2n}M_i = V^{2n}M_i = \Pi V^{2n-1}M_i$, thus Π is injective. This shows, in particular, that the map

$$\Pi\colon \eta_{i-1,n} \longrightarrow \eta_{i,n}$$

induced on the ϕ-invariants is injective. The constructibility follows from the assertions (a) and (b).

The proof of (a): If $i-1$ is not critical, we have by Lemma 6.6 an isomorphism:

$$N_{i-1,n} \cong M_i/V^{2n}M_i$$

such that ϕ corresponds with $V^{-1}\Pi$, thus an isomorphism

$$\eta_{i-1,n} \cong \left(M_i/V^{2n}M_i\right)^{V^{-1}\Pi} \cong \eta_{i,n}.$$

One checks that the above morphism $\eta_{i-1,n} \to \eta_{i,n}$ is induced by Π.

The proof of (b): If $i-1$ is critical, we have by 1 an isomorphism

$$\eta_{i-1,n} \cong \left(M_{i-1}/V^{2n-1}M_i\right)^{V^{-1}\Pi}.$$

Moreover the diagram

$$M_{i-1} \underset{V}{\overset{\Pi}{\rightleftarrows}} M_i \underset{V}{\overset{\Pi}{\rightleftarrows}} M_{i+1}$$

is identified with the diagram

$$\eta_{i-1} \otimes_{\mathcal{O}} \mathcal{W} \underset{\Pi\circ\sigma^{-1}}{\overset{\Pi}{\rightleftarrows}} \eta_i \otimes_{\mathcal{O}} \mathcal{W} \underset{\Pi\circ\sigma^{-1}}{\overset{\Pi}{\rightleftarrows}} \eta_{i-1} \otimes_{\mathcal{O}} \mathcal{W}.$$

The inclusions

$$\Pi V^{2n-1}M_i = V^{2n}M_i \subset \Pi M_{i-1} \subset M_i$$

are σ-invariant and by taking the σ-invariant parts we deduce inclusions

$$\pi^n\eta_i \subset \Pi\eta_{i-1} \subset \eta_i.$$

Thus

$$\eta_{i-1,n} \cong \Pi\eta_{i-1}/\pi^n\eta_i;$$

moreover we have in this case $[\eta_i : \Pi\eta_{i-1}] = 1$, so that

$$\eta_{i,n}/\Pi\eta_{i-1,n} \cong \mathcal{O}/\pi.$$

□

[9] See Proposition 3.6.

By some technical argument, we can deduce a similar proposition for η_j, $j = 0, 1$.

Proposition 8.16. *1. η_i is a smooth π-adic sheaf of free $\mathcal{O}$-modules of rank 2.*

2. η_{i-1} is a constructible π-adic sheaf of free $\mathcal{O}$-modules of rank 2 and $\Pi\colon \eta_{i-1} \longrightarrow \eta_i$ is injective. Moreover:

(a) η_{i-1} is smooth over $S - S_{i-1}$ and Π is an isomorphism over $S - S_{i-1}$.

(b) η_{i-1} is smooth over S_{i-1} and $\eta_i/\Pi\eta_{i-1}$ is a smooth sheaf of free $\mathcal{O}/\pi$-modules of rank 1 over S_{i-1}.

Remark 8.17. There are some points concerning the proposition that may require further clarification, but I do not yet fully understand them. Nevertheless, the previous version appears to be sufficient for the proof of the main theorem.

9 The natural transformation ξ

From this point on, we will only sketch the main arguments and omit the technical details.

9.1 Rigidification

Let B be a $\bar{k}$-algebra, and X a special formal $\mathcal{O}_D$-module of height 4 over B. Write $S = \operatorname{Spec} B$, and for $j \in \{0, 1\}$, let S_j be the closed subscheme of S above which j is critical for X. Recall that Φ is a fixed special formal $\mathcal{O}_D$-module of height 4 over $\bar{k}$.

Definition 9.1. A rigidification of X is a quasi-isogeny $\rho : \Phi_B \to X$ of height 0.

Proposition 9.2. *Suppose that $S = S_i$ for some index i and X is rigidified. Then:*

1. *The sheaf η_i is constant on S.*
2. *The restrictions to $S - S_{i-1}$ and to S_{i-1} of the sheaf η_{i-1} are constant.*
3. *For any rigidification ρ of X, there is a corresponding isomorphism of sheaves*

$$r\colon \underline{K}^2 \xrightarrow{\cong} \eta_0 \otimes_{\mathcal{O}} K$$

such that $[\eta_j|_{S_j} : \Pi^j r\mathcal{O}^2] = j$ for $j \in \{0, 1\}$.

We will use the above proposition without proof.

Proposition 9.3. *Suppose X is rigidified. Then for $j \in \{0, 1\}$:*

1. *The sheaf η_j is a constructible sheaf on S of free $\mathcal{O}$-modules of rank 2.*
2. *The restriction of η_j to S_j is a constant sheaf.*
3. *For any rigidification ρ of X, there is a corresponding isomorphism of sheaves*

$$r\colon \underline{K}^2 \xrightarrow{\cong} \eta_0 \otimes_{\mathcal{O}} K$$

such that $[\eta_j|_{S_j} : \Pi^j r\mathcal{O}^2] = j$.

Remark 9.4. By Proposition 5.8, the constructibility still holds for $B \in \mathrm{Ob}\overline{\mathrm{Nilp}}$.

Proof. The first and second assertions follow immediately by the previous proposition. For the last assertion, we need the following lemma:

Lemma 9.5. *Let I_1 and I_2 be two ideals of B such that $I_1 \cap I_2 = 0$. Let i_1, i_2, and i_{12} be the closed immersions of the subschemes of S defined by I_1, I_2, and $I_1 + I_2$, respectively. Then there is an exact sequence*

$$0 \longrightarrow \eta \longrightarrow i_{1*}i_1^*\eta \oplus i_{2*}i_2^*\eta \longrightarrow i_{12*}i_{12}^*\eta$$

of presheaves on S.

Proof. For every B-algebra B', let $B'_1 = B'/I_1B'$, $B'_2 = B'/I_2B'$, and $B'_{12} = B'/(I_1 + I_2)B'$. The exact sequence

$$0 \longrightarrow B' \longrightarrow B'_1 \times B'_2 \xrightarrow{(a,b)\mapsto a-b} B'_{12} \longrightarrow 0$$

induces an exact sequence of Cartier modules[10]

$$0 \longrightarrow M_{B'} \longrightarrow M_{B'_1} \times M_{B'_2} \longrightarrow M_{B'_{12}} \longrightarrow 0$$

and an exact sequence of modified Cartier modules

$$0 \longrightarrow N(M_{B'}) \longrightarrow N(M_{B'_1}) \times N(M_{B'_2}) \longrightarrow N(M_{B'_{12}}) \longrightarrow 0.$$

Taking the kernel of $1 - \phi$, we obtain an exact sequence

$$0 \longrightarrow \eta(B') \longrightarrow \eta(B'_1) \times \eta(B'_2) \longrightarrow \eta(B'_{12}),$$

which proves the lemma. □

Let $c_j \in B$ be the element corresponding to the morphism $\Pi\colon T_j \to T_{j+1}$ for $j = 0, 1$ after choosing a basis for T_0 and T_1, respectively. Then $(c_0c_1) = (\pi) = 0$. Thus, $\big((c_0) \cap (c_1)\big)^2 \subset (c_0c_1) = 0$. By Proposition 5.8, we may assume $(c_0) \cap (c_1) = 0$. Then let $I_1 = (c_1), I_2 = (c_0)$, which means S_l is the vanishing locus of I_l for $l \in \mathbb{Z}/2\mathbb{Z}$.

Applying the lemma to η and tensoring the sequence by K, we have

$$0 \longrightarrow \eta_0 \otimes_{\mathcal{O}} K \longrightarrow i_{1*}(i_1^*\eta \otimes_{\mathcal{O}} K) \oplus i_{2*}(i_2^*\eta \otimes_{\mathcal{O}} K) \longrightarrow i_{12*}(i_{12}^*\eta \otimes_{\mathcal{O}} K).$$

By the previous proposition, we have $r_l\colon \underline{K^2}_{S_l} \cong i_l^*\eta_0 \otimes_{\mathcal{O}} K$ for $l \in \mathbb{Z}/2\mathbb{Z}$. Then we can glue r_0 and r_1 together to obtain the suitable morphism r if we know the restrictions to $S_0 \cap S_1$ of r_0 and r_1 coincide. We use it as a fact. □

9.2 The natrual transformation ξ

Recall that H is the restriction to the category $\overline{\mathrm{Nilp}}$ of the functor F; and G is Drinfeld's functor[11].

For any $B \in \mathrm{Ob}\overline{\mathrm{Nilp}}$, we define a map $\xi_B\colon G(B) \to H(B)$ by associating a pair (X, ρ) to the quadruple $(\eta_X, T_X, u_X, r_{(X,\rho)})$ where, if M is the Cartier module of X, we have:

1. $\eta_X = \eta_M$ viewed as a sheaf on $\operatorname{Spec} S$;
2. $T_X = \operatorname{Lie} X = M/VM$ viewed as a locally free sheaf on $\operatorname{Spec} B$.

[10] Use Corollary 2.16.
[11] See Definition 4.1.

3. $u_X : \eta_X \to T_X$ is the sheaf morphism.

4. $r_{(X,\rho)} : \underline{K}^2 \to \eta_{X,0}$ is the isomorphism associated to the rigitification of X.

By Proposition 7.9 and Proposition 9.3, ξ defines a natural transformation $G \to H$.

To prove Theorem 4.2, it suffices to show the following.

Theorem 9.6. *The natural transformation ξ is a natural isomorphism.*

Use deformation theory, we can deduce that:

Proposition 9.7. *To show ξ is an isomorphism, it suffices to show that its restriction to the category of $\bar{k}$-algebras is an isomorphism.*

10 Sketch of the final proof

First, let's consider the tangent spaces of G and H.

Definition 10.1. For $Q = G$ or H and $x \in Q(\bar{k})$, the tangent space of Q at x is defined by

$$t_Q(x) = \left\{ x' \in Q(\bar{k}[\varepsilon]) \;\middle|\; x' \mapsto x \text{ under the natural map } Q(\bar{k}[\varepsilon]) \to Q(\bar{k}) \right\}.$$

The name "tangent space" does make sense. If $Q = \mathrm{Hom}_k(\mathrm{Spec}(-), Y)$ for some scheme over $\bar{k}$, Y, and $x \in Y(\bar{k})$, then

$$\begin{aligned} t_Q(x) &= \{ y \in Q(\bar{k}[\varepsilon]/(\varepsilon^2)) : y = x \bmod \varepsilon \} \\ &\cong \mathrm{Hom}_{\bar{k}}(\mathfrak{m}_x/\mathfrak{m}_x^2, \bar{k}) = T_x Y. \end{aligned}$$

We have an important result about their tangent spaces.

Theorem 10.2. *For $x \in G(\bar{k})$, the tangent space $t_G(x)$ is a $\bar{k}$-vector space, and that the tangent map $t_\xi(x) : t_G(x) \to t_H(x)$ induced by ξ is an isomorphism of $\bar{k}$-vector spaces.*

Then, we will give a covering of G.

Definition 10.3. Let $G_{n,m}$ be the subfunctor of G such that $G_{n,m}(B)$ consists of isomorphism classes of pairs (X, ρ) such that $\pi^n \rho : \Phi_B \longrightarrow X$ is an isogeny and there is an isogeny $\beta : X \to \Phi_B$ satisfying $\beta \pi^n \rho = \pi^{n+m}$.

Let $\Phi_B(\pi^{n+m})$ be the kernel of $\pi^{m+n} : \Phi_B \to \Phi_B$, ι_{n+m} the inclusion $\Phi_B(\pi^{n+m}) \hookrightarrow \Phi_B$, and ι'_n the inclusion $\ker(\pi^n \rho) \hookrightarrow \Phi_B$. Let A be the augmented algebra representing $\Phi(\pi^{n+m})$, then A is a q^{4n+4m}-dimensional $\bar{k}$-algebra.

By Proposition 3.4, we have:

$$\begin{aligned} G_{n,m}(B) &= \{ \phi : \ker(\pi^n \phi) \to \Phi_B(\pi^{n+m}) \mid \iota_{n+m} \circ \phi = \iota'_n \} / \sim \\ &\hookrightarrow \{ \tilde{\phi} : A \otimes_{\bar{k}} B \twoheadrightarrow R \mid R \text{ is a locally free } B\text{-algebra of rank } q^{4n} / \} \sim \\ &\hookrightarrow \mathrm{Grass}(q^{4n+4m}, q^{4n+4m} - q^{4n}). \end{aligned}$$

One checks that the inclusions are closed immersions.

Hence $G_{n,m}$ embeds into a Grassmannian. Therefore, we have the following proposition.

Proposition 10.4. *Each $G_{n,m}$ is represented by a projective scheme. If $n \geq n'$ and $m \geq m'$, the inclusion $G_{n',m'} \subseteq G_{n,m}$ is a closed immersion.*

Let Ω be the special fibre of $\widehat{\Omega} \otimes_{\mathcal{O}} \mathcal{O}^{nr}$. By definition, the restriction of H to the category of $\bar{k}$-algebras is representable by Ω.

By the construction of the p-adic upper half plane, Ω consists of copies of $\mathbb{P}^1_{\bar{k}}$ meeting on the k-rational points; at a node the local model is

$$\operatorname{Spec} \bar{k}[x,y]/(xy).$$

The standard affine patches can be written more concretely. Around a node one starts with $\operatorname{Spec} k[x,y]/(xy)$. Remove the k-rational smooth points while retaining the node. If S is the multipicity subset generated by

$$x-1,\ldots,x-(q-1),y-1,\ldots,y-(q-1)),$$

the patch is

$$\operatorname{Spec} S^{-1}\big(k[x,y]/(xy)\big).$$

Gluing such patches according to the Bruhat–Tits tree produces Ω.

Proposition 10.5. *The scheme Ω is separated.*

Proof. Use the following affine criterion[12]: a scheme Y is separated if it has an affine covering $\{U_i\}$ such that each $U_i \cap U_j$ is affine and

$$\mathcal{O}_Y(U_i) \otimes_{\mathbb{Z}} \mathcal{O}_Y(U_j) \longrightarrow \mathcal{O}_Y(U_i \cap U_j)$$

is surjective. For two adjacent standard charts, the intersection is either empty or the axis with all k-rational points removed. One checks carefully that the corresponding ring homomorphism is surjective.

The criterion proves that Ω is separated. □

Let $\xi_{n,m}$ be the restriction of ξ on $G_{n,m}$, then the morphisms

$$\xi_{n,m}(\bar{k})\colon G_{n,m}(\bar{k}) \to H(\bar{k}), \qquad t_{\xi_{n,m}}(x)\colon t_{G_{n.m}}(x) \to t_H(\xi_{n,m}(x)) \text{ for any } x \in G_{n,m}(\bar{k})$$

are injective.

Since $G_{n,m}$ is quasi-compact, by the construction of Ω, there exists an quasi-compact open subset $\Omega_{n,m}$ of finite type such that $\xi_{n,m}$ factors through $\Omega_{n,m}$. Then the induced morphism $\xi_{n,m} : G_{n,m} \to \Omega_{n,m}$ is projective, since a morphism (over $\operatorname{Spec} \bar{k}$)from a projective $\bar{k}$-scheme to a separated $\bar{k}$-scheme is always projective[13].

We will use the following theorem proved in [5].

Theorem 10.6. *Suppose $\varpi\colon X \to Y$ is a projective morphism of finite type $\bar{k}$-schemes that is injective on closed points and injective on tangent vectors at closed points. Then ϖ is a closed embedding.*

[12]See [4]
[13]See [5].

Therefore, $G_{n,m} \to \Omega_{n,m}$ is a closed immersion, which means that $\xi_{n,m}$ is a locally closed immersion with proper source, and hence

$$\xi_{n,m} \colon G_{n,m} \to \Omega$$

is a closed immersion.

Since $G(\bar{k}) = \bigcup_{n,m} G_{n,m}(\bar{k})$ and $\xi(\bar{k})$ is bijective, we can deduce that $\{G_{n,m}\}_{n,m}$ forms a "proper" cover of Ω and hence $\Omega = \bigcup_{n,m} G_{n,m} = G$.

We will explain more clearly. By the construction of Ω, we can find a countable quasi-compact open cover of Ω, $\Omega = \cup_{l \geq 0} \Omega_l$ such that $\Omega_l \subset \Omega_{l+1}$ for any $l \in \mathbb{N}$. Then the closure of Ω_l consists of finitely many projective lines. Since $G_{n,m}$ is an one dimensional closed subscheme of Ω and $G_{n,m}$ is proper over $\bar{k}$, then any projective line contained in Ω will be covered by $G_{n,m}$ for sufficiently big n and m. (Note that the only dimension one subscheme of $\mathbb{P}^1_{\bar{k}}$ which is proper over $\bar{k}$ is $\mathbb{P}^1_{\bar{k}}$ itseslf.) Thus, we could find $n_l, m_l \in \mathbb{N}$ such that $\Omega_l \subset G_{n_l,m_l}$ and $n_l, m_l \to \infty$. Since $G_{n,m}$ is quasi-compact, we can also find $l_{n,m} \in \mathbb{N}$ such that $G_{n,m} \subset \Omega_{l_{n,m}}$. Therefore, ξ is indeed a natural isomorphism.

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